

UNIVERSITY OF Waterloo



Department of Mechanical and Mechatronics Engineering

ME 482: Final Design Report

**Project Sisypheus: An Ethanol-Nitrous Oxide Regeneratively-Cooled Liquid
Rocket Engine for Waterloo Rocketry**

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Summary of Individual Contributions

Matthew Gordon:

Designed and fabricated the fluids GSE, valves, and injectors. Assisted with feedsystem and tank sizing. Conducted the CFD study of the fluid distribution volute with Xavier. Designed and led all testing of components, and the culminating static fire test of the engine. Contributed to all course deliverables.

Joe Dolina:

Electrical design of the new EGSE panel. Oversaw the development of additional EGSE items such as the cart by members of the rocketry team. Designed data acquisition PCB and wrote sections of the report for this and the EGSE.

Xavier Rayes:

Conducted CFD study of fluid distribution manifold with Matthew, 3d printing of the test regen volute, Igniter design and development, Initial vent valve design before descope, heat transfer calculation python script before descoping of test piece, machining of various hardware, report writing.

Joel Godard:

Designed overall vehicle architecture to determine engine requirements, contributed to engine architecture, fabrication of outer tank, small amounts of machining various hardware, report and presentation writing, contributed to component and system level testing/troubleshooting, led project management (Gantt chart, meetings, identifying dependencies in timeline etc.)

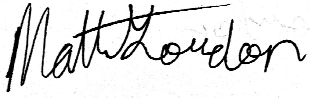
Stefan Arroyo-Cottier:

Wrote the python script that modeled the combustion heat transfer between CC and regen cooling, modelled induced stresses leading to the complete design of the CC (minus the manifold at the bottom) and dealt with its manufacturing. Detail design of the tanks and bulkheads. Manufacturing of various valve components, post machining of CC and injector carrier. Contributed to all course deliverables.

This FYDP team would also like to recognize the work done in support of this project by others not in the project group, including:

- Dan Steinhaur
- Dr. Andrew Milne
- Darien Ng
- Niva Patel
- Les Ankucza
- Ryan Cadeau
- The Waterloo Rocketry Student Design Team

We the undersigned take responsibility for this design

A handwritten signature in black ink that reads "Matthew Gordon". The script is cursive and fluid, with the first name and last name clearly legible.

Matthew Gordon

A handwritten signature in black ink that reads "Joe Dolina". The signature is written in a cursive style, with the first name "Joe" and the last name "Dolina" being distinct.

Joe Dolina

A handwritten signature in black ink that reads "Xavier Rayes". The signature is cursive, with a large initial "X" and the name "Rayes" following in a flowing script.

Xavier Rayes

Joel Godard

A handwritten signature in black ink that reads "Stefan Arroyo-Cottier". The signature is written in a cursive style, with the first name "Stefan" and the last name "Arroyo-Cottier" being clearly legible.

Stefan Arroyo-Cottier

1 Executive Summary

Project Sisyphus is a 7 kN Nitrous Oxide / Ethanol liquid bipropellant engine developed to support low-altitude test flights for the Waterloo Rocketry team. The engine uses regenerative cooling, meaning a heat exchanger with the fuel is formed to prevent the combustion chamber from melting and enable rapid reuse of the engine. This engine also reduces the cost per launch from \$3000 for a single-use solid rocket motor down to \$300 for incremental propellant costs.

This report outlines the design and testing of the Sisyphus engine, which has been fully prototyped and validated, along with the details on the ground support equipment built to support testing of the engine. The Sisyphus engine is now ready for use on a flight vehicle.

Word Count (excluding front matter and appendices): 11,134

2 Problem Background

2.1 Background Information

2.1.1 Waterloo Rocketry

Waterloo Rocketry (WR) is an 80+ person student design team at the University of Waterloo which designs a rocket each year and launches it at the Launch Canada Challenge in Timmins, Ontario. The team's rockets are typically six to eight inches in diameter and 10 to 18 feet long. In 2025, the team launched Aurora, which was an eight inch rocket and flew to 38 000 feet. Based on current goals and team direction, the team is expected to continue to fly eight inch diameter rockets in future years. Notably, for the past two competition rockets, the team has been using a liquid bi-propellant engine architecture (see Sec. 2.1.2 in [1]). This architecture offers many benefits and is commonplace in industry, but is only recently being adopted by teams like WR at the amateur scale.

2.1.2 Amateur Liquid Rocket Designs

Notable among the groups working on amateur liquid rocket designs is Half Cat Rocketry with their Mojave Sphynx vehicle. This is a 4 inch rocket designed to use NOS and almost any hydrocarbon-based fuel. The design is intended to be easy to build, robust, cheap, and safe. They have published a detailed build and design guide which can be accessed here along with many other liquid rocket design best practices on their website [2]. Waterloo Rocketry has also designed two functional liquid rocket engines using ethanol and NOS. The first (Eridium) had a peak thrust of 8.2 kN and a burn time of around five seconds. The second (Eridium V2) has a much higher peak thrust of 12 kN and burn time of seven seconds. Many other student teams are also working on liquid rockets such as MACH (TMU), UBC Rocket Team, and many other US and European university rocket teams. The vast majority of these follow a Half Cat style design using NOS (due to it being self pressurizing), an alcohol based fuel, and ablative cooling of the combustion chamber (CC).

Waterloo Rocketry's 2025 rocket Aurora is shown in Figure 1.

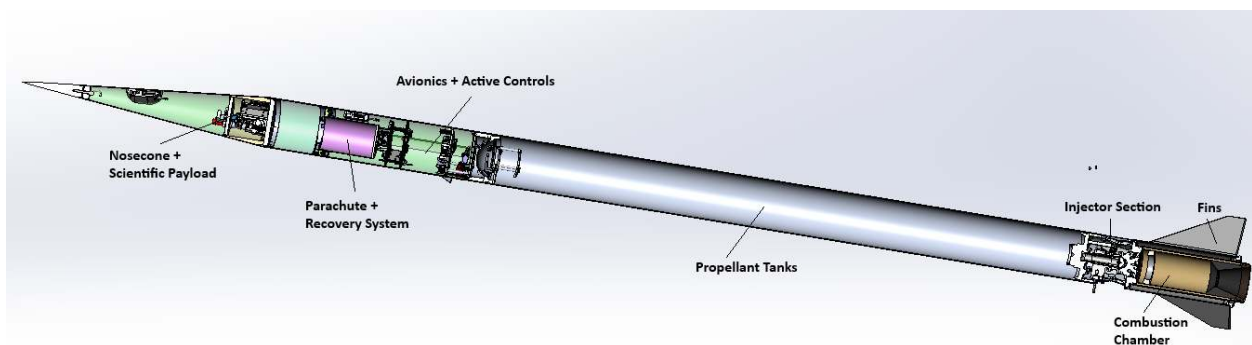


Figure 1: 2025 Waterloo Rocketry Team Vehicle

2.2 Problem Description

Waterloo Rocketry faces a unique challenge of only flying once a year, which limits the amount of representative testing that can occur, especially with respect to subsystems that cannot be easily tested by other means (recovery, payload and controls for example). This challenge is caused largely by a lack of resources and time to build more than one rocket per year. Things like parachute deployments, payload deployments and active flight controls must be tested on the flight vehicle. This has led to instances on the team where subsystems do not work during the competition flight as they could not be sufficiently tested during the year. The team has previously conducted a mid-design cycle launch using a commercial solid motor. These motors, however, are prohibitively expensive for yearly testing, costing upwards of \$3000 for only the motor. These also involve additional logistical complexity as rocket motors are controlled, cannot be transported over the border in the case of US launches, and must be sourced at the launch site. Although a workable solution, this also limits the team to a single launch per test flight campaign.

The aim of this project is to design and build a liquid rocket engine suitable for the Waterloo Rocketry team to conduct yearly test flights that is simple, robust, and reusable, while being comparatively cheaper to commercial solid engines on a per-launch cost basis. From Figure 1, the main scope of this project is therefore the propellant tanks, injector section, and combustion chamber - it is assumed that the upper section including electronics, payloads, and the recovery mechanisms are provided by the customer (the team) as test articles.

2.3 Safety

Safety is a significant consideration for this project. There is the involvement of cryogenic gasses, flammable materials, high pressures as well as the planned attempts for a static fire and other potentially hazardous tests. All of these risks are ones that the team members are familiar with as members of the Rocketry team. As all team members have completed the prerequisite safety training to work in the bay, there is a baseline safety knowledge for every member especially for working with cryogenic materials. To ensure practical safety in the design and particularly testing, the design team will follow all of the same procedures and safeguards that the team uses. This means the usage of the proper PPE, calculations for standoff distances, and the use of dedicated testing spaces. Furthermore, the project team will follow the design guidance and safety regulations put in place by the Launch Canada competition, ensuring compliance on factors of safety and testing plans. The full hazard register for this project was submitted with the initial proposal, and has not changed.

2.4 Society

Space and high altitude are incredibly vast and hard to access frontiers that hold significant scientific and societal value. Many of the designs and research focused on rocketry and space often find their way back into every day life and the use of satellites and space technology is doubtlessly intertwined

with one's daily life. By providing a platform for young student engineers to practice not only their engineering skills but allowing to practice in what can be seen as a difficult field to enter this project is working to directly influence science and technology in Canadian aerospace.

2.5 Professional Ethics

The most relevant ethical aspects for this project are safety and engineering competence. High power rocketry is inherently a dangerous endeavour. By following professional ethics, many steps will be taken to ensure safety is considered and maximized at every step of the project. Due diligence will be performed to ensure every component works reliably and as designed, engineering competence is required to understand potential failure modes and how to mitigate them. Furthermore, ethical behaviour will be followed in ensuring every safety guideline and regulatory recommendation is followed and that no corners are cut.

2.6 Sustainability

The sustainability of this project arises principally from its reuseability, and the use of recyclable materials. Metals such as aluminum and copper are very recyclable and as such, scrap and chips from manufacturing will be able to be recovered. This means that all of the material is being utilized, even if it isn't used on the engine. As this rocket can be used multiple times, it reduces the need for redesign and manufacturing of an engine every year.

2.7 Namesake

In Greek mythology, Sisyphus is a figure cursed to push a large, heavy boulder up a hill again and again for eternity. There is a significant history in the aerospace industry of using Greek or Roman mythological namesakes for rockets (Atlas, Mercury, Saturn, etc), and this re-usable engine is designed to lift heavy payloads up into the sky again and again as part of testflight campaigns, making Sisyphus an apt namesake for this project.

3 Detailed Design

3.1 Combustion Chamber

The combustion chamber is designed to be 3D printed out of CuCrZr, a copper alloy specifically designed for 3D printing. Using copper adds significant margin to the thermal design as it has notably higher heat flux and melting temperature than aluminum, which was baselined during the S25 term. The chamber is designed to be regeneratively cooled, meaning fuel is flowed through the walls of the chamber acting as a heat sink for the thermal energy generated during the combustion process, as shown in Figure 2.

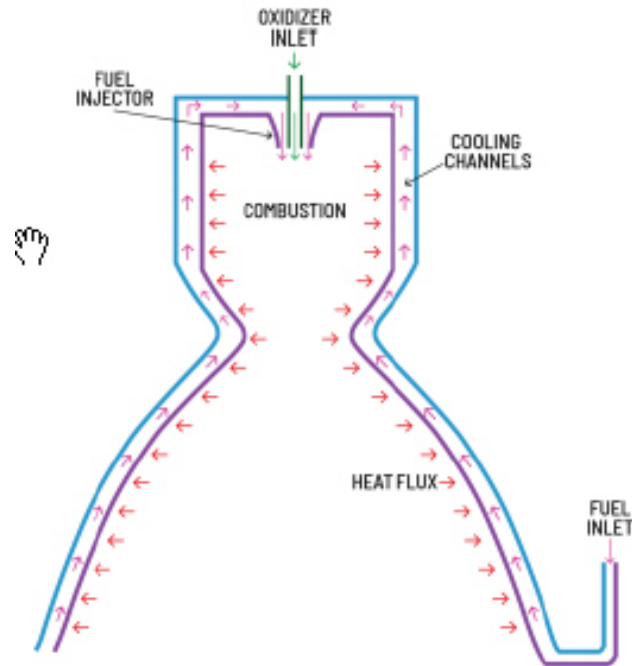


Figure 2: Regeneratively Cooled Combustion Chamber Schematic [3]

The current combustion chamber assembly is shown in Figure 3.

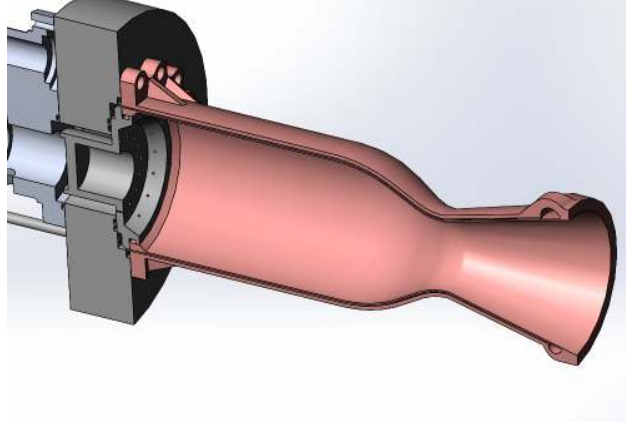


Figure 3: Current Combustion Chamber CAD

The purpose of the combustion chamber is to hold the pressure of combustion and expel the products of the combustion reaction to atmosphere to propel the rocket forward. To properly size this component, the pressures, temperatures and stresses must be known at each location within the chamber. To begin the analysis of the combustion chamber, the use of NASA CEA (Combustion Equilibrium Analysis) was implemented. The feedsystm specifies the use of a 2.7 MPa chamber pressure and a fuel pressure of 5.5 MPa at an OF ratio of 2 to run as cool as possible while maintaining relatively high specific impulse (ISP, also known as efficiency). CEA gives the gas properties of the resulting products such as the specific heat ratio, the temperature of combustion in the chamber, at the throat, and at the nozzle exit. Using a desired thrust of 7 kN, the same information is used in a software called RPA [4] (Rocket Propulsion Analysis) to export a curve of a engine geometry which will provide the desired thrust at desired conditions. Using the isentropic flow equations, the mach number can be determined at each point from the ratio of areas between each point and the throat. The adiabatic flame temperature can then be determined at every point along the length of the combustion chamber in the same manner.

Once the flame temperature at each location is known, a thermal circuit can be implemented to solve for the wall and ethanol temperatures. This circuit implements use of convective and conductive relationships to model how heat flux moves from the heat of combustion into the ethanol to increase the temperature of the fuel, removing it from the combustion chamber walls. The thermal circuit which defines this problem can be seen in Figure 4.

The thermal circuit is defined with a convective resistance applied to the combustion gases, a conduction resistance applied through the copper and a final convective resistance into the ethanol. The convective resistance through the combustion gases is defined by the Bartz equation seen in Figure 5.

The conductive resistance is defined by the thickness of the copper, the thermal conductivity and the area of interest. Finally, the convective resistance through the ethanol is defined by the Gnielinski

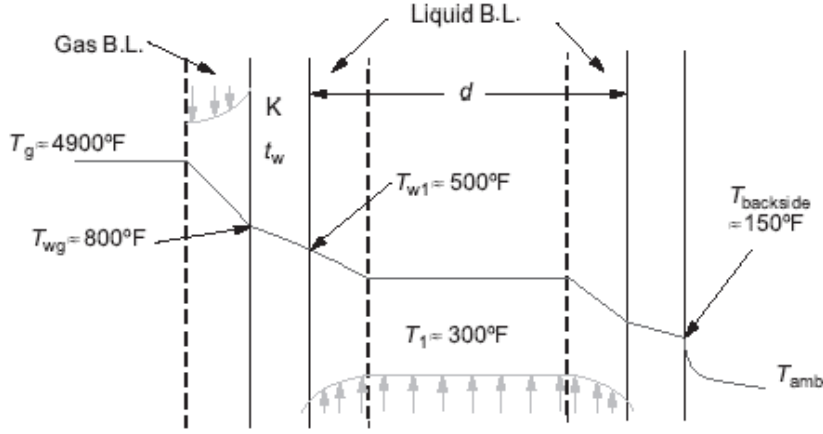


Figure 4: Thermal Circuit of the Regen System [5]

$$h_g = \frac{0.026}{D_*^{0.2}} \left(\frac{\mu^{0.2} c_p}{Pr^{0.6}} \right) \left(\frac{p_c}{c^*} \right)^{0.8} \left(\frac{D_*}{r_c} \right)^{0.1} \left(\frac{A_t}{A} \right)^{0.9} \sigma$$

- h_g is the [heat transfer coefficient](#) from the hot combustion gas to the chamber/nozzle interior wall (W/m²/K)
- D_* is the diameter of the chamber throat (m)
- μ is the [dynamic viscosity](#) of the combustion gas (Pa s)
- c_p is the [specific heat capacity](#) of the combustion gas (J/kg/K)
- Pr is the [Prandtl number](#) of the combustion gas
- p_c is the chamber pressure (Pa)
- c^* is the [characteristic velocity](#) of the combustion reaction (m/s)
- r_c is the [radius of curvature](#) of the throat wall (m)
- A_t is the cross sectional area of the chamber throat (m²)
- A is the cross sectional area of the chamber/nozzle at the point of interest (m²)
- σ is a dimensionless parameter accounting for variation of gas properties across the [boundary layer](#):

$$\sigma = \frac{1}{\left(\frac{1}{2} \frac{T_w}{T_c} \left(1 + \frac{\gamma-1}{2} M^2 \right) + \frac{1}{2} \right)^{0.8-\omega/5} \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\omega/5}}$$

- T_w is the temperature of the chamber/nozzle interior wall (K)
- T_c is the chamber temperature (K)
- γ is the [ratio of specific heats](#) of the combustion gas
- M is the [Mach number](#) of the combustion gas at the point of interest
- ω is the exponent of a viscosity-temperature power law for the combustion gas. A common estimation is 0.6. See [temperature dependence of viscosity](#).

Figure 5: Bartz Equation and Variables [6]

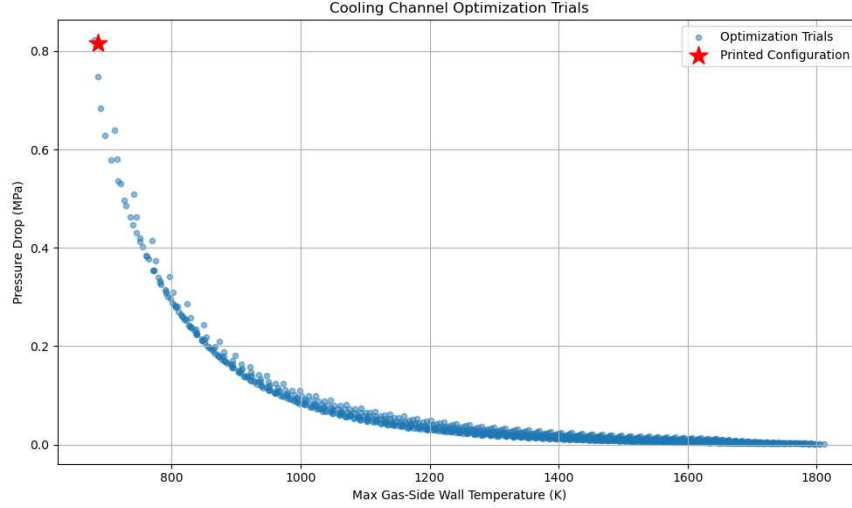


Figure 6: Channel Optimization Results

correlation seen below, a correlation which defines the Nusselt number to find the convective heat transfer coefficient in the channel.

To solve for the temperature at steady state, all of this information can be used in an iterative process. The Bartz correlation is dependent on wall temperature. Therefore, to have an original guess for the heat flux, a wall temperature is guessed. Given that the flame temperature is known and the initial ethanol temperature can be assumed to be ambient, the heat flux through the convective resistance can be determined. Then, using the derived heat flux, the inner wall temperature can be determined across the conductive resistance. Finally, given this new inner wall temperature, a heat flux into the ethanol can be found over the last convective resistance. This new heat flux can be compared to the previous heat flux. If they are not within a tolerance, Newtons method, or any other iterative approach can be used to make a new wall guess until the heat fluxes agree. Using this value of Q on a given volume of ethanol, $Q = mc\Delta T$ can be used to determine the new ethanol temperature when leaving the defined area. This new ethanol temperature can then be used in the next thermal circuit. From this, the flame temperature, inner and outer wall temperatures, and the ethanol temperature can be determined at each location along the combustion chamber.

A function was implemented to estimate the frictional pressure losses from the small channels using the Darcy friction factor and typical major losses formulas. A maximum acceptable pressure loss value was defined by the injector section, that is, the minimum amount of pressure the injectors could accept. With a known pressure limit and a function that provided the maximum wall temperature for a given channel geometry the channel sizing could be conducted. With a smallest channel size defined by the limits of 3D printing technology with a resolution of 1mm, the smallest feasible channel is 1mmx1mm. With a smallest channel size defined an upper bound of channel size was defined to be roughly 5mmx5mm and the solution space was descritized and iterated though, finding the pressure loss and maximum wall temperature at every point and plotting in Figure 6.

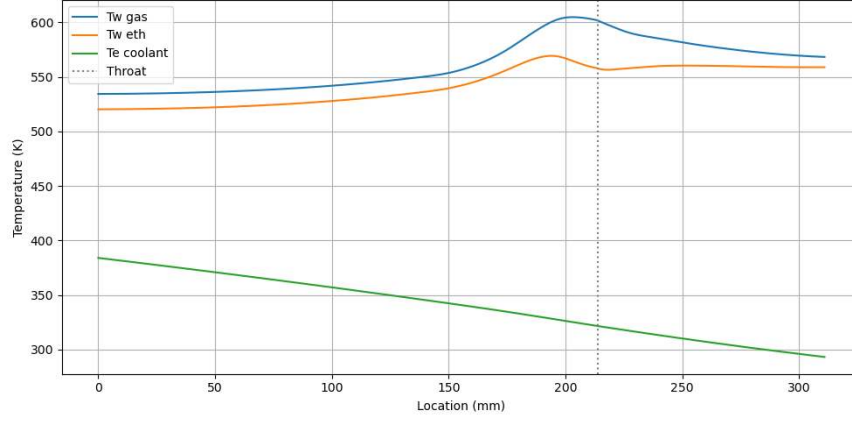


Figure 7: Wall Temperatures at Steady State

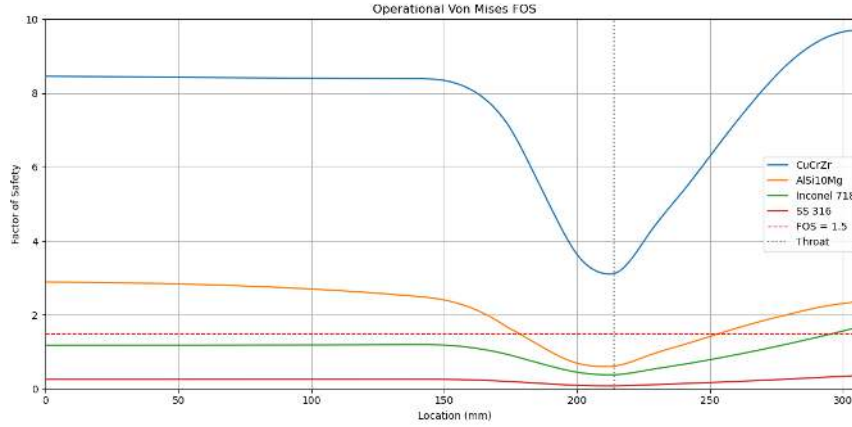


Figure 8: Von-Mises Stress of Different Materials

This plot demonstrates how there exists an optimal curve on the left most boundary of the point cloud. This curve defines the lowest possible pressure drop for every wall temperature. Given every point is below the acceptable max pressure drop of 1 MPa the coolest configuration was chosen (ie. the smallest possible channels manufacturable).

The temperature distribution of the engine being fired at an ambient temperature of 20C is show in Figure 7.

As a part of the material selection Stainless Steel, Inconel, AlSi10Mg and CuCrZr were considered for their printability and heritage in regeneratively cooled engines. The temperatures calculated as per the method above were converted into thermal stress (from the temperature gradient) and pressure stresses (from the pressures in the channels and chamber). Temperature dependent yield strength values were found and the Von-Mises stress was calculated at each point as shown in Figure 8. From this plot it is clear that CuCrZr is the only feasible material with a FOS on yield of above 3 at steady state.

To ensure printability no overhang greater than 45 degrees was designed. This inherently required some post machining, namely the following 3 steps:

1. Tapping ports for PTs and ethanol inlets
2. Flycutting the top sealing face for a clean and flat finish
3. Drilling out and reverse counter-boring the holes for fasteners

3.2 Injectors

The injectors are the component responsible for controlling the flow and mixing of propellants entering the combustion chamber. The specific constraints ('shall' requirements) and criteria ('should' requirements) are presented below, along with their current validation status at the time of writing.

Injector Requirement	Rationale	Status
The injectors shall mix NOS and ethanol at a mass ratio of 2:1	Ratio based on thrust and chamber temperature targets	Actual O:F ratio of 2.2:1 based on static fire testing, 10% variation and a shift towards being ox rich is acceptable, so this requirement is met
The pressure upstream of the injector shall be at least 540 psia	This is 20% higher than the chamber pressure of 450 psia, required to prevent 'chugging' instability	Pressure drop exceeded 30% in all testing
The injectors shall include a volume in which to mount the igniter	This keeps the igniter as close to the mixing point as possible, minimizing routing work	Included by design, but potentially contributed to melting, requirement to be revisited
The injector shall include single redundancy on any inter-propellant seals	This means, for example, a back-up o-ring to ensure no mixing of propellants occurs in the injector manifold	All inter-propellant seals are doubled, Requirement is met
The injector shall be sized using fully annealed aluminum 6061	Aluminum is the easiest material to source and machine for the team, and is lightweight. Annealed 6061 is used for strength calculations to ensure failure does not occur after repeated engine firings which may anneal the aluminum	Annealed aluminum used in all sizing math

The injectors should be machinable in the E5 student shop (ie on a manual machine)	This allows prototyping and manufacturing to be done relatively cheaply and efficiently, and allows for tuning or future changes to improve performance without significant time or expense	Injectors machined by hand in approximately 12 man hours
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Injectors control the flow of propellant using pressure drop across an orifice or similar flow channel, setting the mass flow rate entering into the combustion chamber. The mass flow rate depends on the size of the orifice, and the pressure drop. 'Chugging' instability is a type of instability in a rocket engine in which chamber pressure and propellant flow rate form a feedback cycle. When the pressure drop across the injectors is small, a slight variation in chamber pressure has a disproportional impact on the pressure drop, resulting in a notable change in mass flow rate. Because chamber pressure is dependant on the rate at which propellant is burning (due to a fixed nozzle size), this then starts a feedback loop - if chamber pressure drops, mass flow will increase, causing chamber pressure to rise, causing mass flow to drop, etc. A 20% difference between chamber pressure and injector upstream pressure is generally sufficient to avoid this [7].

Many variations of injector designs exist. A selection are shown in 9.

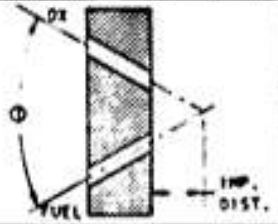
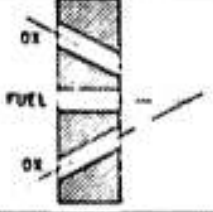
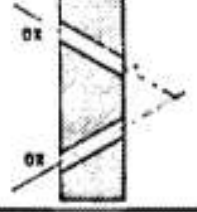
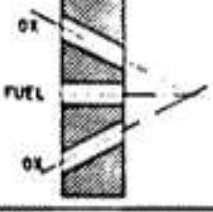
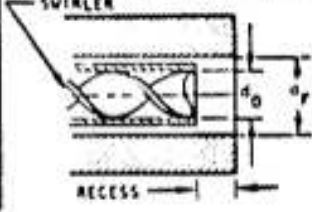
ELEMENT DESIGNATION	ELEMENT CONFIGURATION (FLOW DIRECTION)
UNLIKE DOUBLET (1 ON 1) 	
UNLIKE TRIPLET (2 ON 1) 	
UNLIKE QUADLET (2 ON 2) 	
UNLIKE PENTAD (4 ON 1) 	
CONCENTRIC TUBE (WITH SWIRLER)	

Figure 9: Injector Designs [8]

Of these designs, the primary ones initially considered for this project were the unlike-triplet and the unlike-doublet, as they require only drilling holes. Unlike doublet style injectors are ideal for mixing two liquids, with the two impinging streams being one oxidizer and one fuel. Unlike triplet, on the other hand, is ideal for mixing a liquid propellant with a gaseous propellant, as the two outer liquid jets impinge and atomize, and then mix with the central gaseous stream, providing

much better overall atomization and mixing than would occur if streams of liquid and gas were directly impinged (due to the significantly higher momentum of the liquid jet).

Note that while the overall oxidizer to fuel ratio is 2, the injector is designed such that the jets are sized with an oxidizer to fuel ratio of 2.5 (slightly more oxidizer-rich, and therefore easier to ignite). The additional fuel mass flow is injected through small holes known as 'film cooling holes', which are straight holes around the outer diameter of the chamber that inject fluid along the outer wall, providing a fuel-rich and therefore cooler boundary layer.

The parameters of the first revision injector design are as follows:

Parameter	Value
Injector Type	Unlike Dobulet
Number of Elements (Fuel + Ox Orifice Pairs)	16
Fuel Orifice Diameter [mm]	1.25
Oxidizer Orifice Diameter [mm]	1.78
Number of Fuel Film Cooling Holes	8
Film Cooling Orifice Diameter [mm]	0.85

3.2.1 Fluid Models

Sizing the injector orifices requires a way to predict the mass flow rate that will occur through an orifice at a given pressure drop. For the ethanol fuel, which is a single-phase incompressible (SPI) liquid at the operating conditions, the mass flow $\dot{m}$ can simply be done using the sharp-edged orifice equation derived from Bernoulli's principle, which is dependant on the fluid density ρ , the orifice area A , and the pressure drop over the orifice ΔP .

$$\dot{m}_{SPI} = C_d * A * \sqrt{2 * \rho * \Delta P}$$

In this equation C_d is an empirical parameter known as the discharge coefficient, which accounts for the effects of frictional and minor losses, and ranges from 0 to 1.

The oxidizer used for the Sisyphus engine, Nitrous Oxide, on the other hand, is much more complicated to model. At ambient conditions, Nitrous Oxide is a vapour, however at pressures starting around 700 psia (at 20°C) NOS becomes a liquid. NOS has a very high vapour pressure, which means that for amateur rocketry designs it is typically stored as a saturated liquid and the vapour pressure alone is used to push the NOS out of the propellant tank. However, given NOS is stored as a saturated liquid any pressure drop will cause vapour formation.

There are two primary models which can be used to model this two-phase flow through an orifice. The first, the Homogeneous Equilibrium Model (HEM), assumes isentropic flow and thermal

equilibrium, resulting in the equation below.

$$\dot{m}_{HEM} = C_d A \rho_2 \sqrt{2(h_1 - h_2)}$$

The HEM model is a capped model, meaning that once it reaches the 'critical' or maximum mass flow rate it is assumed that this mass flow remains constant even if the pressure drop is further increased (despite the mathematical model predicting a decrease past this point). However, this results in the model tending to under predict the mass flow rate for NOS where the pressure drop is above that required for the critical HEM mass flow. To solve this, the Dyer model was proposed which averages out the mass flow rates predicted by the SPI and HEM models.

$$\dot{m}_{Dyer} = \frac{\kappa}{1 + \kappa} \dot{m}_{SPI} + \frac{1}{1 + \kappa} \dot{m}_{HEM}$$

Where

$$\kappa = \sqrt{\frac{P_1 - P_2}{P_{vap} - P_2}}$$

which represents a ratio between the vapour bubble growth and residence times in the orifice [9].

A representative plot of the predicted mass flow rates per unit area for the SPI, HEM, and Dyer models at varying pressure drops is shown in Figure 10. Note for this plot the tank pressure is held constant and chamber pressure was varied, meaning a lower chamber pressure results in a higher mass flow rate.

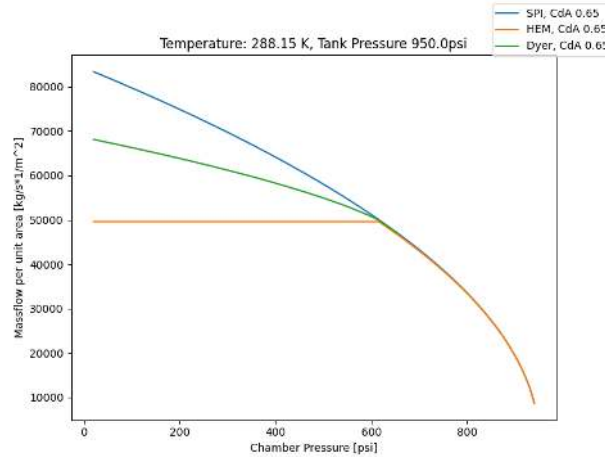


Figure 10: Mass Flow Models

Note that for this specific engine, since the NOS is subcooled by pressurizing it above its vapour pressure with gaseous N₂, the NOS should start as a subcooled liquid (the extent of subcooling will depend on the initial temperature) and so the expected mass flow rate will be somewhere between the Dyer and SPI predictions.

3.2.2 Conceptual designs

The initial two designs considered are shown below, with the unlike triplet in Figure 11 and the unlike doublet in Figure 12.

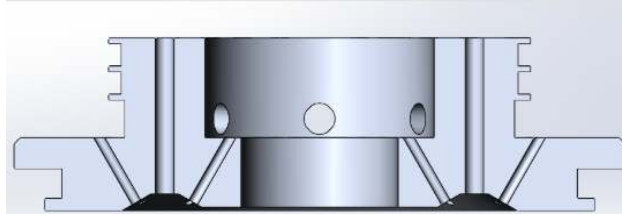


Figure 11: Unlike Triplet Concept

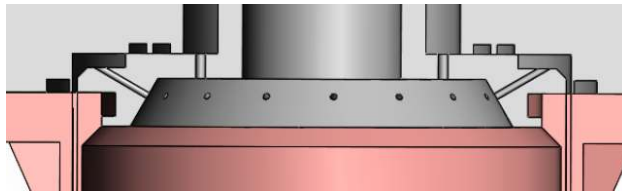


Figure 12: Unlike Doublet Concept

In both cases the fuel enters from the outer diameter after coming up through the regen cooling channels, and the oxidizer enters from the top through the vertical orifices. The unlike triplet injector uses long oxidizer orifices to increase the residence time, and therefore the amount of vapour generated, resulting in what should be a largely vapour jet as preferred by the unlike triplet design. On the other hand, the unlike doublet design uses short oxidizer holes to minimize residence time and therefore vapour generation, keeping the NOS behaviour closer to SPI (due to the sub-cooling) and resulting in a liquid jet entering the chamber and impinging with the single angled fuel jet. For both injectors, the impingement angle between the jets was set to 60° per recommended practice in SP-8089 [8].

The unlike doublet design was ultimately selected for two main reasons:

1. The unlike doublet is significantly easier to machine, as there is no need for the cross-holes to transfer fuel to the inner ring present on the unlike triplet design. These cross holes could also add pressure drop and result in non-uniformity of the two fuel jet velocities, which would bias combustion towards the centre of the chamber and potentially melt the injector faceplate.
2. Testing of the unlike doublet is much easier due to the SPI-like behaviour expected in this design, as it can be done at lower pressures with water for development tests instead of requiring a 2-phase fluid (often liquid Carbon Dioxide) to properly simulate the 2-phase behaviour of NOS and get reasonable data.

The finished injectors are shown in Figure 13.



Figure 13: Machined Injectors

3.2.3 Testing

Initial testing was conducted using water to verify the flow pattern and mixing on a single-element test article, as shown in Figure 14, where the two streams represent the oxidizer and fuel respectively.



Figure 14: Injector Mixing Testing

Testing of both the oxidizer and fuel side were also conducted independently, using safer simulated propellants - carbon dioxide to simulate NOS and water to simulate ethanol. The results of this testing are as follows:

Parameter	Value
Fuel Side $C_d A$ [m^2]	0.000014255
Oxidizer Side $C_d A$ [m^2]	0.000028457

The oxidizer side $C_d A$ value was calculated using the Dyer method. In both cases, and using readings from a load cell monitoring the mass of the oxidizer run tank and a pressure sensor in the oxidizer tank to determine the pressure drop and mass flow rate. It was assumed that the pressure in the chamber was atmospheric for this calculation. The data from the oxidizer side test is presented in Figure 15 , and Figure 16 shows the CO2 flow.

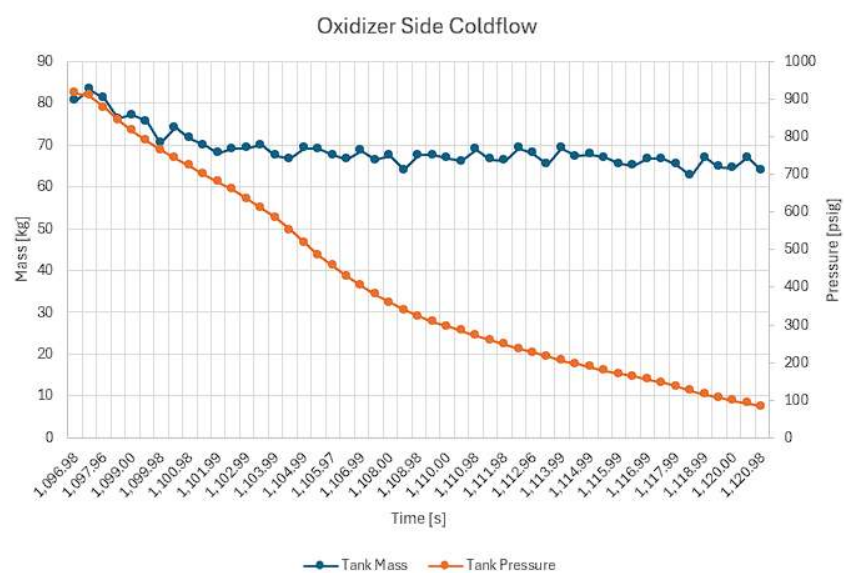


Figure 15: Oxidizer Coldflow Data



Figure 16: Oxidizer Side Coldflow

A linear fit was performed on the mass data over the $\tilde{6}$ s where liquid CO₂ was flowed, resulting in a mass flow rate of 2.656 kg/s as shown in Figure 17.

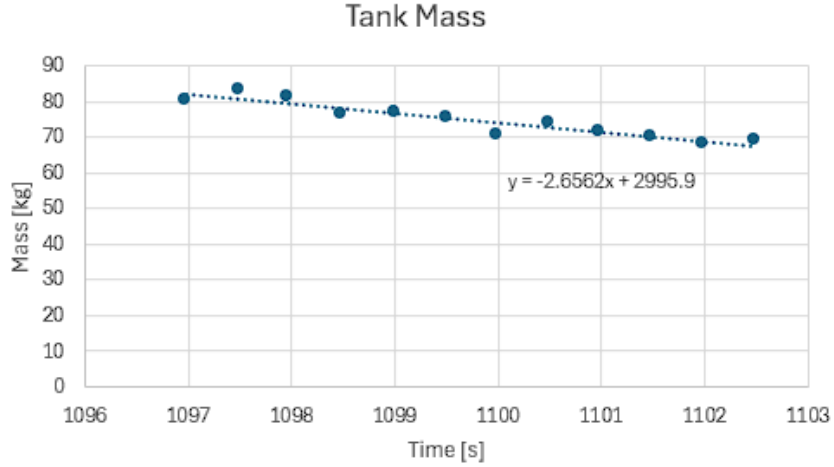


Figure 17: Oxidizer Side Coldflow Linear Fit

On the fuel side, a test was conducted by filling the fuel tank fully with water (which has a known volume and therefore allows total mass to be calculated without a load cell), pressurizing the tank to 200 psia, and timing how long it took for all the water to be expelled, giving a mass flow rate and pressure to use in the SPI equations. In the test, 10.2 kg of water was expelled in 11.63 seconds.

Note that due to the configuration of the fuel feed lines, specifically how the regenerative cooling channels feed directly into the fuel side of the injector, the fuel flow test had to be conducted with the chamber installed, and so the $C_d A$ value for fuel includes the cooling channels. Additionally, the team was unable to verify the flow pattern on the finished injectors as with the chamber installed (require to flow fuel) the mixing could not be observed.

Based on the data from the independent coldflows, the overall O:F ratio for these injectors under flight conditions was predicted to be 2.18.

3.2.4 Static Fire, Melting, and Redesign

During the first static fire test of the Sisyphus engine, both thrust and chamber pressure data were unfortunately lost due to technical and electrical errors, meaning the true O:F ratio could not be conclusively determined. However, using tank pressures and total propellant mass flow rates, a script was used to attempt to estimate the actual O:F ratio, which was estimated as 2.16. The O:F ratio will be conclusively determined in future tests, and the methodology for determining this value is discussed later in the testing section.

There were two injector-specific learnings from the static fire test. The first was that the igniter cap and a portion of the igniter housing melted during the burn, as shown in Figure 18, with Figure 19 as a reference for what the face initially looked like - note the igniter cap, colored in red, completely melted.



Figure 18: Injector Face Post-Static Fire

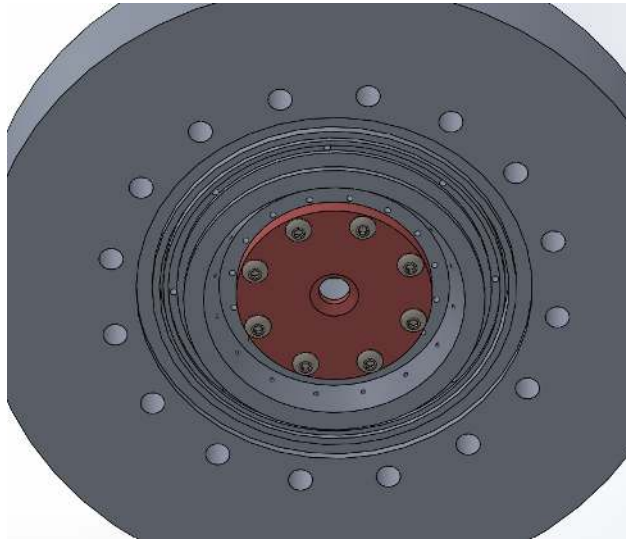


Figure 19: CAD of Injector Face

It are a couple potential contributing causes to this melting. First, in the nominal design the mixing zone of propellants is 0.5mm away from the edge of the igniter cap, meaning combustion would have been very close to the part that melted, resulting in a high convective and radiative heat loading. As an exercise, treating the burning gas and igniter cap as two parallel disks and assuming an emissivity of 1 for the hot gas at 2300K and 0.65 for the aluminum, and assuming the 32g cap was thermally isolated from the rest of the injector (a conservative assumption), the

predicted temperature rise over a 5s burn was 380 K, demonstrating the high radiative heat loads on the injector face.

The second observation was high-frequency combustion instability, appearing as an audible high-pitched shriek in the footage of the test. In brief, high frequency combustion instability is an indicator of poorer or inconsistent mixing, which is suspected to be due to higher-than-anticipated vapor formation in the oxidizer jet, causing the impinging fuel jet to blast right through it and not fully mix.

To solve both of these issues, the injectors have been redesigned, as shown in Figure 20. The key design changes are:

- Igniter moved from inside the injector to hanging from the face (not shown), and previous igniter cavity filled, increasing the injector part mass from 190g to 285g. Retention method has not yet been finalized, but is expected to be an ABS pin or twist lock that will both allow the igniter to be installed up the throat of the chamber (without removing the chamber from the injector) and is designed to melt away during the burn.
- Oxidizer holes moved closer to the centre, which reduces the 'dead zone' in the middle receiving the peak radiative heating, moves convective cooling due to oxidizer flow closer to the melted zone, and pushes the mixing location further from the injector face.
- Deleted the igniter cap, which along with moving the oxidizer holes closer in moves the fire face 17x further from the injector face. Temperature rise for a 5s burn due to radiative heating predicted as 33K over a 5s burn for this new design.
- Changed the design to a 'like quadlet' - using the same number and size of holes, but now with pairs of oxidizer and fuel jets impinging on themselves at 60 degrees (ensuring equal momentum between jets and full atomization regardless of vapour content), with the resulting droplet fans then intersecting at 60 degrees. This is shown by the sketched lines in Figure 20.

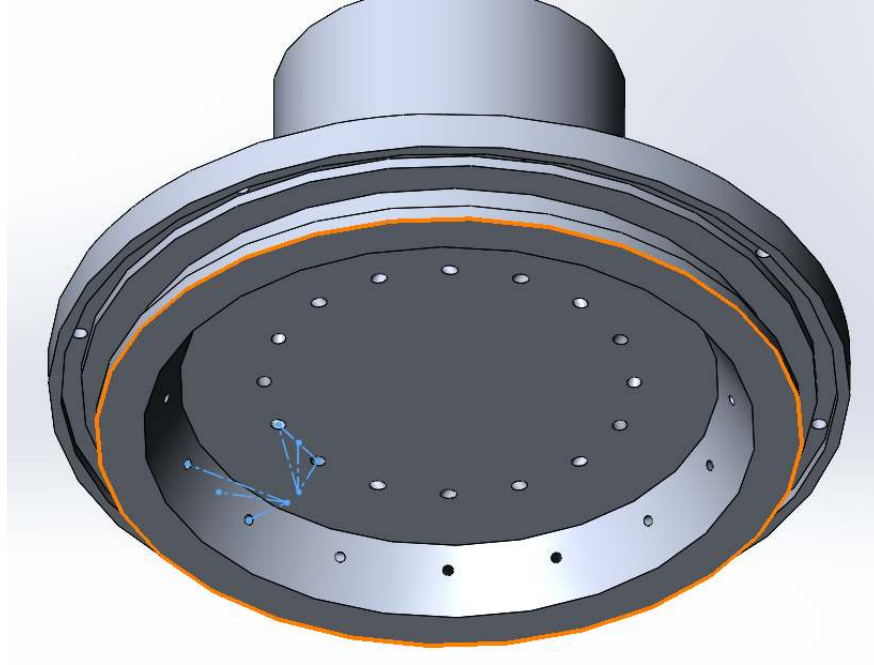


Figure 20: CAD of New Injector

3.3 Feedsystem

The feedsystem for the Sisyphus engine consists of the propellant storage tanks, valves, sensors, and other fluid connections between the tanks and the main combustion chamber. The feedsystem is designed such that it is 'flight-ready', despite an actual flight test of a full vehicle being out of scope for this capstone project. In general, this means the system is vertically integrated within an 8" diameter cylindrical envelope (to match current team hardware) and is somewhat mass-optimized. Additional remaining work required to integrate the feedsystem into a flight vehicle is discussed at the end of this report.

The main injector valves responsible for controlling flow from the tanks into the engine are identical to coaxial sleeve valve designs used previously by Waterloo Rocketry, slightly resized for the specific flow rates. Given the design of these valves is extensively documented in other reports such as [10], and for the sake of focusing this report on the new designs for the Sisyphus engine, these valves will not be discussed in detail.

Additionally, a full Plumbing and Instrumentation Diagram (P&ID) of the engine in its test configuration is provided in the appendices.

3.3.1 Tanks

The propellant tanks are concentric, with the outer tank made from 8" outer diameter 0.25" thick aluminum 6061 tube, and the inner tank made from 3.5" diameter 0.065" thick drawn aluminum 6061 tube. Machined aluminum 6061 closures are bolted onto the tanks and use radial o-ring seals.

Imperial stock sizes are used due to local availability. Concentric tanks are used currently by the Waterloo Rocketry team.

In the concentric tanks shown in Figure 21, the fuel is stored in the inner, and oxidizer is stored in the annulus between the inner and outer tanks. A piston sits in the inner tank (hence the need for a drawn inner tube, to ensure tight tolerances and roundness for the full length) and allows both the fuel and oxidizer to be pressurized by the gaseous volume (also known as ullage) in the oxidizer side of the tank. The piston is bottomed out before filling fuel to ensure no air is trapped on the fuel side of the piston, which keeps the fuel and oxidizer pressures as close as possible through the entire engine burn.



Figure 21: Final Propellant Tank Design

In current Waterloo Rocketry team designs, the oxidizer ullage volume is pressurized by heating the oxidizer (nitrous oxide, or NOS) until its vapour pressure reaches the desired maximum expected operating pressure (MEOP), usually around 700-800 psia. This is an efficient solution, and does a decent job of limiting tank pressure drop during a burn (as liquid leaves the tank, some liquid oxidizer flashes to vapour to refill the empty volume and maintain tank pressure), but has two notable downsides:

- During winter testing, heating up the nitrous oxide to the target temperature can take up to 2 hours
- Hotter nitrous oxide is less dense, requiring larger and heavier tanks to store

In the design used for the Sisyphus engine, the tank is pressurized instead by an external nitrogen supply, which allows a higher maximum expected operating pressure (MEOP) of 1000 psia (very close to the point at which nitrous oxide becomes supercritical). This MEOP was selected to give additional margin on pressure drop over the small regenerative cooling channels. Nitrogen pressurization also allows for significantly faster testing, as nitrogen can be filled in minutes, however

it also requires larger tanks, as more nitrogen is not added as liquid drains from the tank, and so a larger initial ullage volume is needed to limit how much the tank pressure drops over the course of the burn.

Initially, a design which featured separate fuel and oxidizer tanks, each with their own nitrogen-filled ullage volume, was proposed, as shown in Figure 22

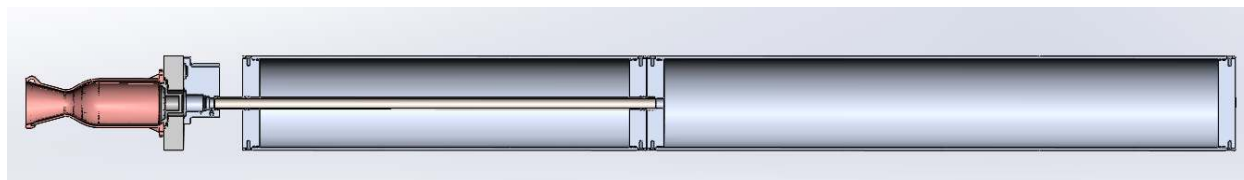


Figure 22: Rejected Propellant Tank Design

However, this design was rejected in favour of the concentric design for the following reasons:

- In fall 2025, the Tripoli Rocketry Association (TRA), which regulates most amateur launch ranges, began insuring liquid-fuelled rockets only if they were pressurized by nitrous oxide vapour, and not rockets with tank pressurized by nitrogen or any other gas. TRA insurance means rockets can fly at a licensed range without needing to go through a formal Closure of Airspace process (CoA) required for uninsured rockets, which takes up to 6 months. For launching this engine, the Waterloo Rocketry team would therefore now almost certainly opt to use NOS vapour for pressurization. Both external nitrogen and heated NOS pressurization are possible in a concentric tank, meaning a single concentric tank could be obtained and used for ground testing using nitrogen in the winter, and eventually launch using NOS vapour pressure.
- The inner drawn tank is the same size as the one used on the Waterloo Rocketry team's large competition rockets (albeit 6' long instead of 8'). In 2025 the Waterloo Rocketry team received a damaged tank which had a dent about 1' from the end, meaning it was not usable for the team's competition rocket, but had enough good material to be cut down and used for this engine. Given the team received a free replacement but got to keep the damaged tank, moving to concentric tanks reduced tank cost by approximately \$1500 CAD.
- In the concentric tank setup, the fuel and oxidizer pressure are directly linked by the piston, keeping the ratio between the feed rates of the two propellants constant, whereas with separate tanks the feed rates may drift without very careful design. This is very desirable given this engine is designed to operate at an O:F ratio of 2, and higher O:F ratio mixtures burn hotter and could result in mixing.

The key design parameters of the final tank design are tabulated below:

Parameter	Value
-----------	-------

Outer Tank OD [mm]	203.2
Outer Tank ID [mm]	190.5
Inner Tank OD [mm]	88.9
Inner Tank ID [mm]	85.6
Tank Length [m]	1960
Initial Pressure [MPa]	6.9
Pressure at End of Burn [MPa]	4.8
Maximum Oxidizer Fill [kg]	33.5
Maximum Fuel Fill [kg]	7.75

The maximum fuel fill is determined by when the piston hits the top of the tank, and the max oxidizer fill is based on the volume that can be filled leaving 10% ullage (recommended to prevent thermal expansion of the propellants from overstressing the tank as could occur if it was 100% full of liquid). Note that the max fuel fill realistically limits how much oxidizer can be filled to 15.5 kg, as to prevent melting of the chamber when fuel runs out the burn should end, meaning no more oxidizer should be present.

The limiting stress on the tanks is tear-out of the M8 fasteners used to retain the closures on either end, and with 30 holes on either side a factor of safety of 2 is attained on this load case.

The tank was sized to achieve the target starting and ending pressures using a 4s burn, and assuming the nitrogen added to the ullage expands as an ideal gas, and assuming an initial amount of NOS (or carbon dioxide) vapour is in the ullage but no more vapour is formed as fluid is expelled. This behavior was validated during a flow test of the tank using CO₂, as shown in the plot in Figure 23. Both the adiabatic and ideal gas expansions are plotted, with the real data (solid blue trace) matching the ideal gas expansion model best.

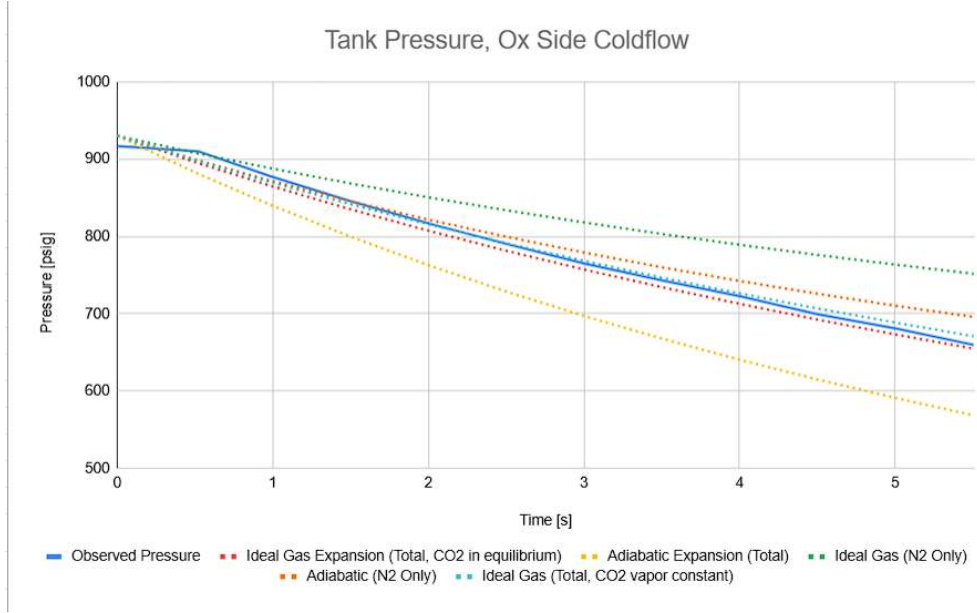


Figure 23: Coldflow Tank Pressure Data

As a note, cases where no additional vapour is generated and where additional vapour is generated were considered (the dashed blue and dashed red traces, respectively). In this case there was little difference between the two, and both match the real data well. The assumption of no additional vapour generation is significantly easier to model, and so it is recommended that this method be used when sizing similar tanks in future.

3.4 Fluid Distribution Manifold

Regenerative cooling in a rocket engine involves flowing the fuel through thin channels in the walls of the chamber before injection, allowing the transfer of heat from the up to 2500K combustion gases into the fuel. This keeps the temperature of the metal combustion chamber walls below their melting point and allows the engine to continue firing for as long as fuel can be supplied.

In order to accomplish this even flow distribution to every regen channel, a manifold was designed to be integrated into the combustion chamber. The regenerative cooling channels have been previously sized to be 3mm x 1mm rectangles, with a total of 84 channels around the perimeter of the chamber. Thermal analysis on the chamber was conducted with fuel (ethanol) flowing at 6 m/s into the inlet of the channels at a pressure of 5.5 MPa and determined this flow results in acceptable thermal margins. Additionally, the fuel flowrate has been sized to be 1.25 kg/s based on the required engine thrust of 7kN and specified oxidizer to fuel ratio of 2. In order to evaluate the performance of the volute, the velocity profile normal to the surface over the outlet flow area was plotted to verify the velocity is no less than 5 % below the target of 6 m/s at any location.

In order to accomplish this even flow distribution, a series of designs were created on which CFD was conducted. This began with a normal inlet, followed by a tangential inlet, followed by a lofted tangential inlet and finally, a dual lofted tangential inlet. An example of the progression of these designs can be seen below in Figure 24.

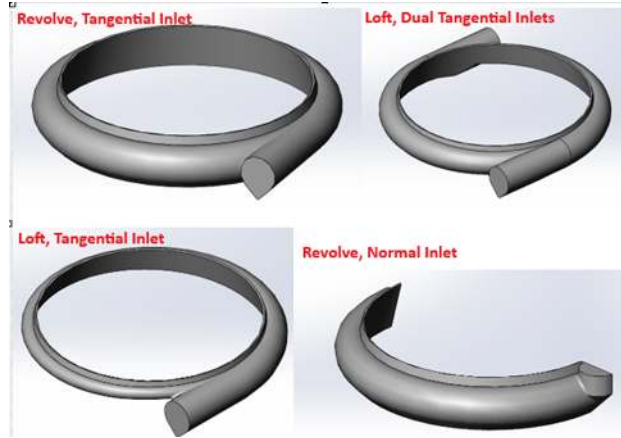


Figure 24: Progression of Fluid Distribution Manifold Designs

In these cases, an identical hydraulic diameter area was used instead of the equivalent channels and was used as the output to easily plot velocity along the manifold to ensure the required fluid distribution. A mesh was then generated which utilized 5 inflation layers and a growth rate of 1.2. In this mesh the walls were set as boundaries and used an element sizing of $1e-04\text{m}$ applied to the outlet face. This mesh can be seen below in Figure 25.

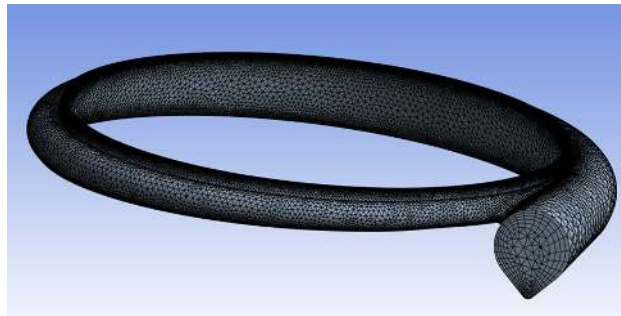


Figure 25: CFD Mesh

These same mesh conditions were applied to all four geometries in Figure 7. Once these mesh conditions were applied, the boundary conditions of the flow were set as the following.

Based on these conditions, and the densities of ethanol and water respectively for different representative simulations, the following velocity fields and their corresponding outlet velocity profiles were solved. These results are shown in Figures 26 through 31.

Table 5: CFD Boundary Conditions

Parameter	Value
Inlet Mass Flow	1.25 kg/s
Inlet Temperature	25°C
Inlet Turbulence	Medium Intensity (5%)
Outlet Pressure	5.5 MPa (relative), averaged over outlet
Convergence Criteria	RMS Error < 1e-4

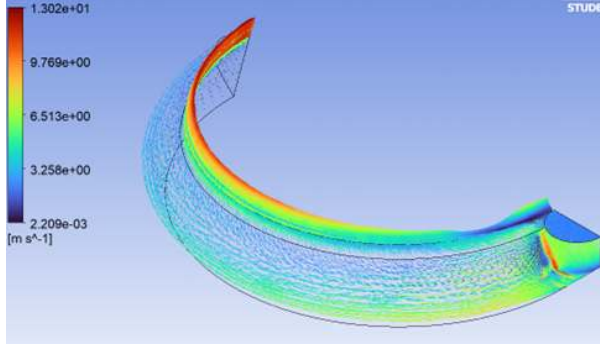


Figure 26: Normal Velocity Solution Field

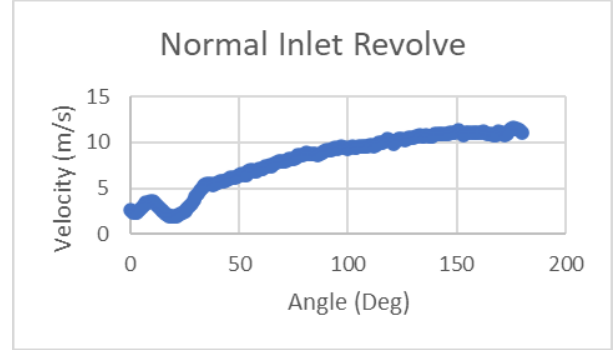


Figure 27: Normal Velocity Outlet Plot

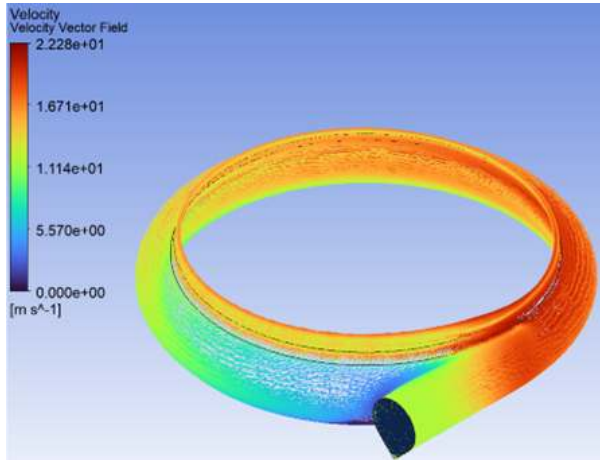


Figure 28: Normal Velocity Solution Field

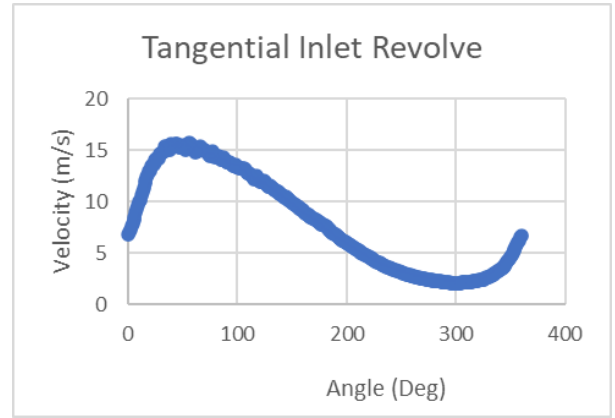


Figure 29: Normal Velocity Outlet Plot

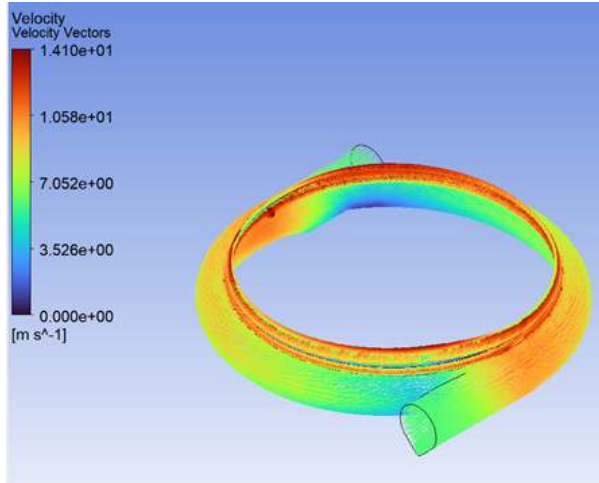


Figure 30: Dual Inlet Velocity Solution Field



Figure 31: Dual Inlet Velocity Outlet Plot

3.4.1 Fluid Distribution Manifold Validation

To validate the CFD model and ensure adequate flow to all channels, a scale replica of the volute geometry was printed out of resin and tapped to allow for a 1/4 NPT fitting to allow pressurized water to flow through the volute geometry. An image from this testing can be seen in Figure 32 below.

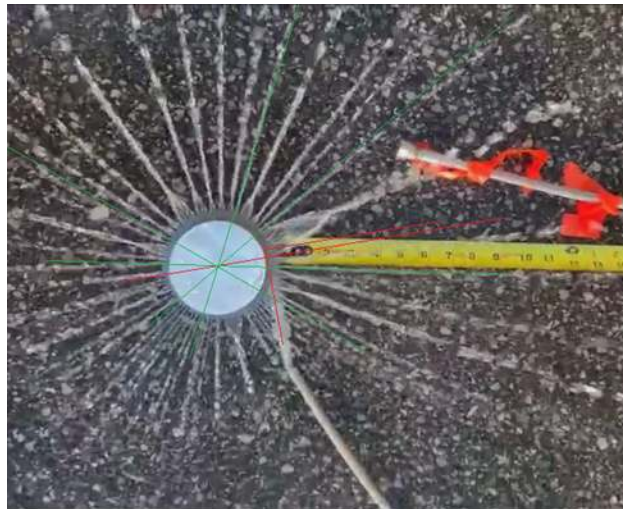


Figure 32: Volute Water Test

Conducting the same simulation with the same boundary conditions as seen during this test creates a representative CFD case. The outlet velocities from the test were then measured and plotted against each other which can be seen in Figure 33.

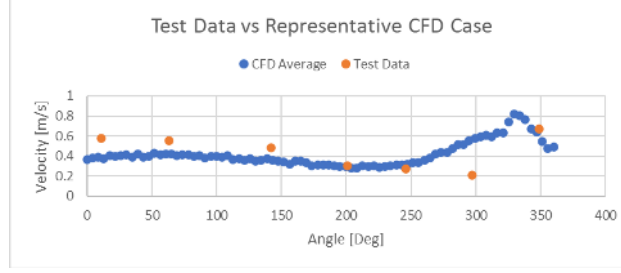


Figure 33: Comparison of Test and CFD Data

Grid Convergence Studies were also completed which validated the CFD model, but the above plot further cemented our understanding of the geometry's weak points. This test allowed for a discussion about the pressure drop across the volute which was less uniform than would be ideal. It was for this reason that the double inlet tangential volute was decided to be the final manifold geometry.

3.5 Igniters

In order to light the rocket engine, a spark must be created to complete the fire triangle of fuel, oxidizer and a spark. This is done with a component known as an igniter. There are two main methods of achieving this. It can be done with a torch igniter or an integrated solid rocket motor. A torch igniter is essentially a small liquid engine which has a plume large enough to light the main engine mixture. However, this has relatively large amounts of complexity and development effort in comparison with the solid rocket motor option. Additionally, potassium nitrate/resin solid mixtures had been previously used on the team however were only used under atmospheric pressure. When enclosed in the injector plate, pressure buildup causes different burn characteristics which could be characterized for future use by the team. In general, a large mass flow is desired into the engine as a large mass flow would allow for a larger plume and therefore a higher likelihood of a successful engine light. The regression rate of these pressurized solid rocket motors is often characterized by Vieille's Law seen below. Where a is the burn coefficient, n is the pressure coefficient (which are both empirically determined and cannot be theoretically derived), and p is the chamber pressure.

$$r = aP^n \quad (1)$$

When sizing the first iteration of the igniter, it was assumed that, like most solid fuel grains seen in papers, the value of n would be positive and less than 1. Therefore, to get the highest mass flow, the pressure would need to be as high as possible, meaning the vent size should be as small as possible. However, after firing multiple iterations of the igniters, it was clear that the value of n was negative and less than 1. This phenomenon had not been accounted for and indicated that the specific mix of potassium nitrate and resin being used was a "Mesa" propellant. A Mesa propellant is a rare propellant where as the pressure builds, a "liquid slag" layer develops around the burning

surface, inhibiting heat transfer, thereby decreasing the regression rate, decreasing mass flow. To account for this, the throat size was increased from 4 mm to 9 mm. This resulted in the following test and corresponding curve fit.

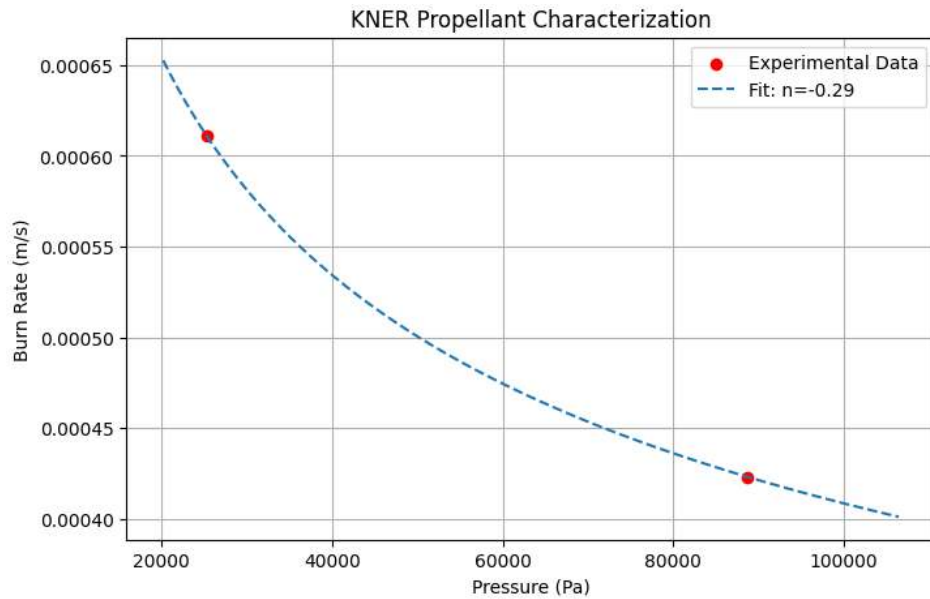


Figure 34: KNER Propellant Characterization



Figure 35: Igniter Test

This is a rather qualitative test but from this, there is 5 seconds of strong burn and 18 seconds of weaker, yet still relatively strong burn. From this, 3 more tests were done to ensure consistent ignition, which allowed the design to be validated.

Constraints	Rationale
Shall provide necessary heat and spark into the combustion chamber to allow for a self sustaining combustion reaction.	The engine must be able to ignite.
Shall be remotely operated.	The engine must be able to be ignited remotely for the safety of the operations crew.
The igniter shall be reusable with a new RNX puck / black powder insert.	With the purpose of this project being a reusable engine, the igniter is a large part of the overarching goal. As such, it should be able to be reused with minimal re-fabrication (3D printing and casting RNX).
Shall integrate into the injector bulkhead.	The igniter need to deliver heat to the combustion chamber at start-up, and the injector bulkhead is the closest component to the chamber with easy fluid pass-through.

Criteria	Rationale
Igniter burn should be at least 3 seconds	This should allow margin for ignition, heat build-up, and valve opening while the igniter is on.
The igniter should direct the ignition flame in the towards the location of propellant mixing.	If the igniter does not send spark up to the correct area, the chamber runs the risk of not lighting (not desired).

Note that from an operational perspective this design requires a well trained operator to recognize the igniter is burning properly and open the injector valves in a narrow time window. This creates a significant risk of engine hard starts or failed ignition but was deemed tolerable due to previous Waterloo Rocketry experience with this type of system. It was considered to remedy this with a pressure sensor in the igniter volume and automatic software opening of the valves but due to packaging issues in the igniter area and the risk of developing and testing an auto-sequence system this was deemed even higher risk than the chosen solution.

3.6 EGSE

The previous design of the rocket-side EGSE was split into four subsystems: Ground Side Power Distribution (GSPD), Remote Launch Control System (RLCS), Data Acquisition (DAQ), and the router box. Each existed inside its own weatherproof enclosure, and they were connected together via cables for power and data transmission.

This design failed to meet a number of the requirements outlined in section 7.1.3. Notably it was not actually waterproof, was susceptible to AC power failure, and was difficult to maintain and transport. Over the Fall term, the design of the EGSE was revised in collaboration with Waterloo

Rocketry to meet these requirements. The revised design is detailed in [1]. In addition to the four subsystems outlined in that report, the network video recorder (NVR) and cameras used by the team to collect test and flight footage was also integrated into the new system. The NVR was acquired after the construction of the original EGSE, and had previously been connected directly to 120 VAC from the generator and protected from the environment with a haphazard and unreliable setup.

3.6.1 Final Design and Verification

No additional design work occurred prior to the construction and assembly of the panel. During assembly, the placement of the outer panel connectors was modified to better fit the as-assembled internals. Also, as some components arrived late, 3D printed mounts were designed and printed in parallel during assembly. Finally, the fuse block on the main 12V bus was expanded to four fuses in total to have each output circuit draw 100 W or less, necessary for potential CSA certification.

The new EGSE has been used successfully at four integrated tests: two cold flows and two static fires between the main and FYDP teams. Table 8 outlines each system requirement, and how it was validated (and if any deviations are present). The fully assembled panel internals are shown in Figure 36.

Table 8: Verification

Constraints	Verification
Shall support up to 10 motorized ball valves or igniters in any combination	System currently supports 7 actuators and 1 igniter, which have been verified. Not enough functional actuator PCBs or ports were available, and this configuration met the needs of both teams for testing. WR will expand the relay channels as necessary for engine tests and flight later this year.
Shall support data acquisition from at least 12 4-wire sensors	The new system has 15 ports which support 1 4-wire sensor or 2 2-wire sensors each. The bank of 6 ports currently in use has been verified.
Shall provide at least 2 parallel interfaces to the ground side CAN network supporting at least 4 pins, 2A current each	The custom sensor PCB was descope for the static tests, making this interface unnecessary for the FYDP tests. All internal hardware except wiring has been installed.
Shall be remotely operable from at least 1000 ft away	WR operated the system from approx. 700 ft. away during their static fire test, and the wireless antenna link has been previously shown to work over 1000+ feet.
All connectors carrying over 25V shall be of appropriate polarity given direction of power flow	Guaranteed by design

No SRAD electronics shall directly interface with 120VAC input power	Guaranteed by design
Actuator power shall be controlled with a dedicated lockout key switch	RLCS design unchanged from old system, key switch has been verified to work
System should be water and debris-proof (meet IPx6 or NEMA 3x/4x/6x ratings as applicable)	The four integrated tests were conducted in harsh outdoor, winter environments. A wash-down test of the panel has not yet been conducted, as the risk of failure was considered higher than that at test. A wash-down test will be conducted prior to the team's competition
AC power (generator) failure should not de-power the system	The generator failed during the team's static fire, and the system successfully failed over to battery power, allowing the test to continue
All connectors for a given system (sensors, actuators, power feed) should be unique	Guaranteed by design
Battery backup should be passively recharged from 120VAC	Selected UPS charges the backup battery
All electrical ground support equipment should be assembled in the field by non-electrical personnel in 30 min or less	Electrical personnel were present at every test, and assembly was not explicitly tested. Given the current state of labeling, it is unlikely this requirement would be met
All electrical ground support equipment should transportable safely by one person	The system has been safely transported by multiple individuals

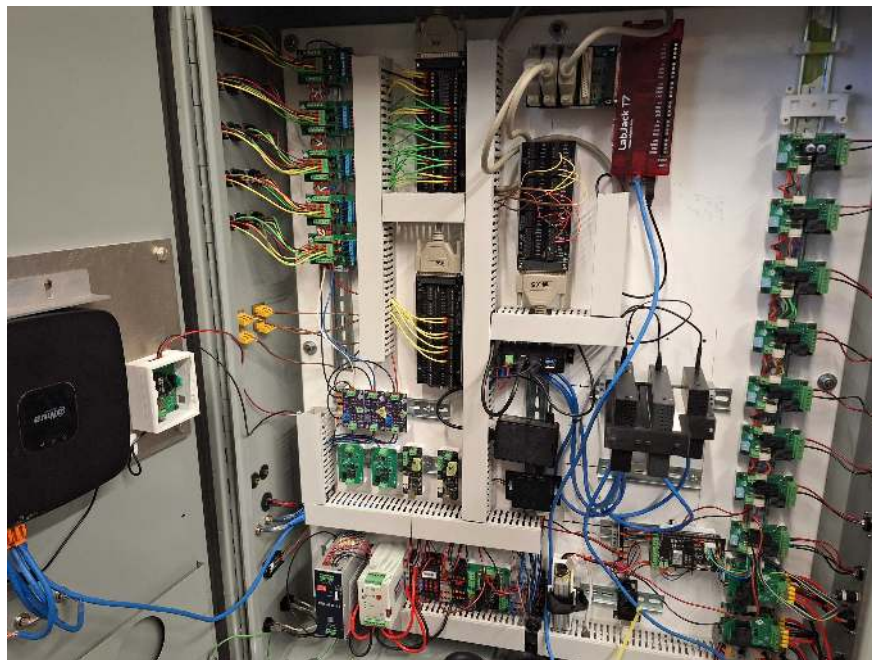


Figure 36: New EGSE panel

3.7 Fluids GSE

The previously existing fluids GSE was a vertical panel-mounted design, as shown in Figure 37.



Figure 37: Existing Fluids GSE

Having fixed fill system plumbing has been a major asset, however the current design has several issues. First, the remotely operated ball valves are electrically driven with no mechanical fail state, meaning if power is lost the valves will not move and could trap propellants in the tank. To deal with this, the current design has rope-actuated manual valves (so that an operator can safely depressurize the system from farther away), however due to the panel being tall and thin pulling on the ropes to open the manual valves have tipped the entire panel over in the past. The panel is also not portable on uneven terrain. Small wheels were installed on the base which allow the panel to be rolled well on concrete or pavement, but on dirt or gravel the wheels jam up and the panel needs to be carried, which is very awkward given its form factor. Finally, the current panel can only fill two propellants - nitrous oxide (or CO₂) and low pressure nitrogen gas.

A new 'fill cart' was built for this project with several upgrades, as listed below:

- Pneumatically actuated remote valves with spring returns from JFC and Assured Automation, allowing mechanical fail states even in the event of power loss
- A low-to-the ground form factor so pulling on rope-actuated valves will not tip the cart over
- A high pressure nitrogen supply line (1000 psia), as required to pressurize the Sisyphus engine tanks
- A carbon dioxide supply line, including a filter (due to CO₂ cylinders being notably dirtier than nitrous oxide cylinders), allowing for no plumbing changes for coldflow tests and the

ability to purge the engine with CO₂ after a burn and extinguish any fires on or around the test stand.

- 4 carry handles for improved portability

In future, there is the possibility that larger wheels could be added to the cart. This system may also be directly mounted on the Waterloo Rocketry team's new test trailer for transport, by fastening through the strut channel on the bottom. The new system is shown in Figures 39 and , and the cart is included in the P&Id of the engine in test configuration found in the appendices.

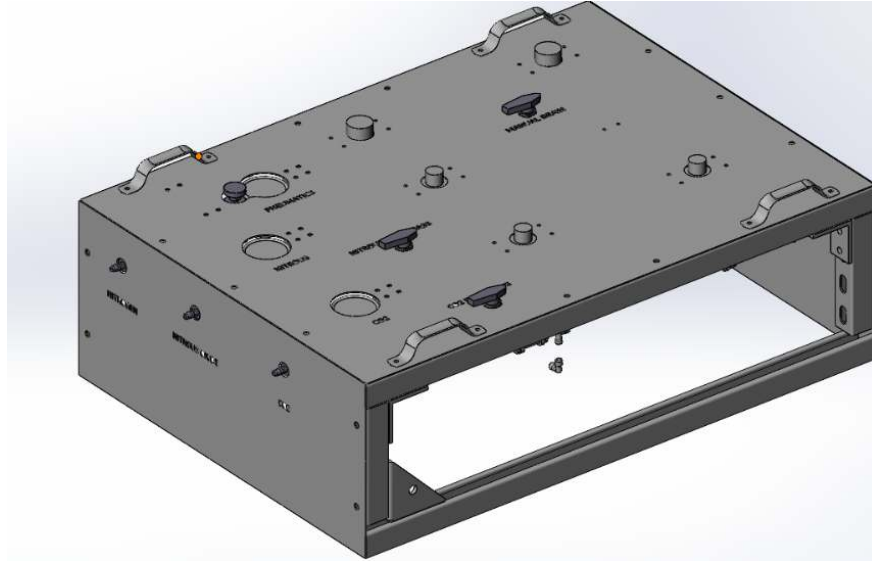


Figure 38: New Fluids GSE CAD



Figure 39: New Fluids GSE

At the time of writing, this new fluids GSE has successfully operated during 4 tests, and the ability to manage filling nitrogen, nitrous oxide, and carbon dioxide simultaneously has been verified. A pneumatics reservoir was added to the low pressure nitrogen side after initial testing showed the pneumatics regulator could not sustain pressure when one of the large quarter-turn actuators was activated.

4 Testing and Validation

The primary methods of testing and validating design requirements are summarized below. More comprehensive test data can be found in the appendices.

4.1 Hydrostatic Testing

Hydrostatic testing was conducted on all custom designed components intended to carry pressurized fluids. This testing involves filling the component with water, and applying pressure using a hand pump. Hydrostatic testing is conducted to limit the damage should component failure occur, as the energy stored in a pressurized incompressible fluid is orders of magnitude lower than the energy stored in a compressible fluid.

All custom components for this project were tested up to 1.5x the maximum pressure they are expected to see in operation (MEOP) and no leaks or deformation was observed. For valves, hydrostatic testing also included testing of the valve's ability to open and close (and seal) under pressure.

Note that this testing is not intended to validate the 'designed to FOS 2' requirement on all fluidics components, as that requirement is specific to the design and was validated by analysis. This testing was instead purely a risk mitigation and workmanship screening test before proceeding to higher risk testing with compressible or combustible fluids.

4.2 Coldflow Testing

Coldflow testing uses 'simulated propellants', or fluids that behave in a similar manner to the intended propellants but are not combustible. For nitrous oxide, carbon dioxide is a suitable simulant, and for ethanol, water can be used. Coldflow testing involves running through all procedures for filling and conditioning the propellants using the simulated propellants, allowing for the functional requirements of the ground support equipment and feedsystem to be validated. Coldflow testing was also used to validate flow rates through the injectors as discussed previously.

Coldflow testing of the oxidizer side was conducted at the Waterloo Rocketry team's James Downey Way test site due to the higher pressures and risk associated with this test, and testing of the fuel side with water was conducted in the SDC washbay due to the low pressures used. Data from the coldflow tests is available in the appendices.

4.3 Static Fire Testing

Static Fire testing was the final verification test performed on the Sisyphus engine for this project. In this test, the engine was secured to a fixed structure (the test stand) at the Waterloo Rocketry team's test site. After fully assembling the system and conducting leak checks at low pressure (200 psi), the test site was evacuated and propellants were filled remotely. The engine was then ignited and fired successfully. Many sensors were connected to the engine to obtain the data required to validate requirements, including 10 pressure sensors, 3 thermocouples, and 2 load cells (one on the propellant tank and one measuring the thrust of the engine). The full data recorded from this test, along with the operating procedures used, are linked in the appendices.

Unfortunately, due to electrical and setup errors, the data from the thrust loadcell and the chamber pressure sensor were lost. However, using the flow data obtained from coldflows, the pressure and temperature of propellants in the tank, and the overall mass flow rate of propellants (2.8 kg/s as recorded by the tank load cell), a script was written to iteratively estimate the chamber pressure. This script worked by calculating the mass flow rates of oxidizer and fuel using the pressure drop (known tank pressure minus the assumed chamber pressure) and the discharge coefficients and validated fluid models, and then comparing the estimated total massflow to the recorded value and adjusting the chamber pressure guess accordingly. A convergence criterion of error less than 0.01 kg/s was used, resulting in the estimated thrust and chamber pressure data as plotted below in Figures 40 and 41.

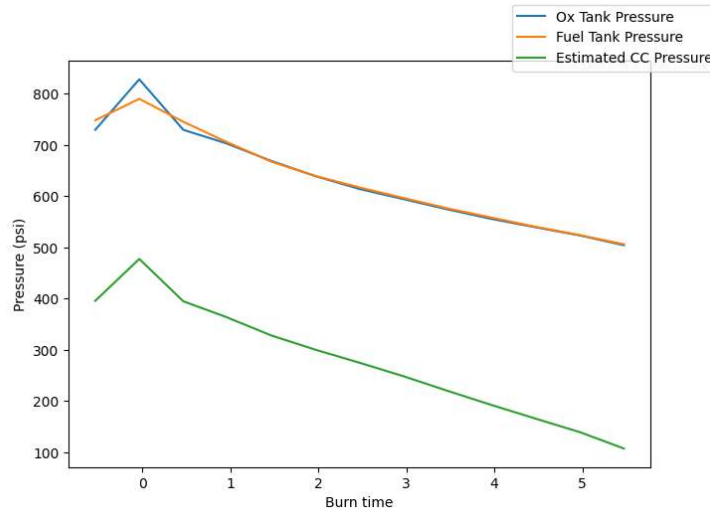


Figure 40: Estimated Chamber Pressure

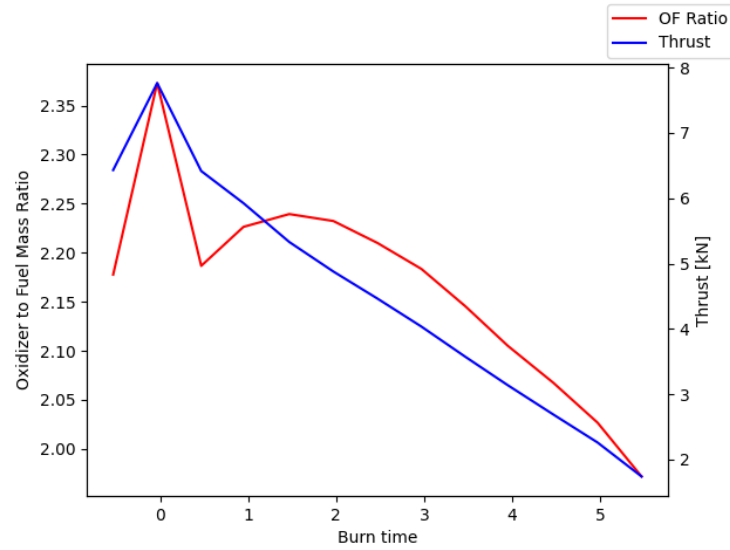


Figure 41: Estimated Thrust and OF Ratio

The static fire was also recorded from multiple angles, a frame of the recording is shown in Figure 42, and the full video is linked in the appendices.



Figure 42: Static Fire

The data from static fire validated all outstanding requirements, except for the requirement that the engine be re-fireable within 1 hour, which was not possible in this test due to the previously discussed melting of the injector face. However, an initial attempt at the static fire test had to be aborted due to a battery powering the ignition coil not being installed, and the system was successfully recycled (propellants fully drained, fixes made, and propellants refilled) within 30 minutes, meaning that as long as the injector redesign works and replacing the igniter takes less than 30 minutes this requirement will be met during the additional testing scheduled to take place in mid-April.

One additional note about the static fire test that was conducted is that the tank pressure was slightly lower than desired due to a partially-empty nitrogen bottle being used, resulting in the total mass flow being lower than desired, however the initial requirements for the engine were still met and extra care will be taken to use a brand new nitrogen bottle for the testing in mid-April.

5 Project Management

5.1 End of Project Deliverables

The end of project deliverables (to the Waterloo Rocketry team) are as follows:

- Full engine and feedsystem CAD and drawings
- Fully manufactured and tested engine that meets aforementioned requirements
- Operations and assembly procedures
- Testing data
- Design documentation

With the submission of this report, all items have now been delivered to the Waterloo Rocketry team. Note that a detailed line-by-line verification of all requirements is presented in the Final Design Review presentation, linked in the appendices.

5.2 Work Breakdown Structure

Each group member was expected to work about 8 hours a week with more as required to hit project deadlines. The actual estimated work breakdown and key tasks are tabulated below:

Group Member	Average Hours Per Week	Key Tasks
Stefan Arroyo-Cottier	25-30	Combustion chamber design, machining, other detailed mechanical design
Joel Godard	10-15	Project management, systems design, manufacturing support, mechanical design
Matthew Gordon	25-30	Feedsystem and valve mechanical design, injector design, test design, manufacturing
Xavier Rayes	10-15	Ignition system design and testing, manufacturing support, mechanical design
Joe Dolina	25-30	All electrical and software design, manufacturing, and testing

A full breakdown of the hours anticipated per task is presented in the appendices. Based on the project finishing on time and internal review of timelines, this breakdown is accurate to the actual time spent, meaning initial estimates were accurate.

5.3 Project Schedule

The project schedule is outlined in the Gantt Chart that can be found in the appendix of this report. The Gantt chart was very useful for planning. The Gantt chart was frequently referred to, and was extended and updated in a team meeting at the start of term. The winter term update was very well executed, as little to no deviation from the schedule occurred during the term. The schedule for winter term was done with the primary goal of having footage of the static fire test in time for symposium, and so the schedule was built around static firing on March 7th/8th, giving 1.5 weeks of schedule margin.

Static fire ended up being on February 28th due to the Waterloo Rocketry team slipping with testing of their competition engine to the weekend of the 7th/8th at the last minute, however this was a potential issue the fydp group had foreseen, and the team was doing well enough that they were able to put in some longer hours and pull the schedule left to hit the earlier test date.

5.4 Organizational Chart & Roles

The organizational chart can be found below:

Project Role	Member	Roles and Responsibilities
Project Member	Stefan Arroyo-Cottier	Combustion Chamber Design
Project Member	Joel Godard	Mechanical & Systems Design
Project Member	Matthew Gordon	Feedsystem Mechanical & Electrical Design
Project Member	Xavier Rayes	Igniter & Combustion Chamber Design
Project Member	Joe Dolina	Electrical and Software Design
Faculty Advisor	Dr. Andrew Milne	Overall Project Guidance
Course Instructors	Profs. Peter Teertstra, H. J. Kwon	Provide Deliverable Feedback
Client	Waterloo Rocketry Team	Provide Design Input
Sponsors	Assorted	Provide Funding

These roles are all related via the organizational chart below:

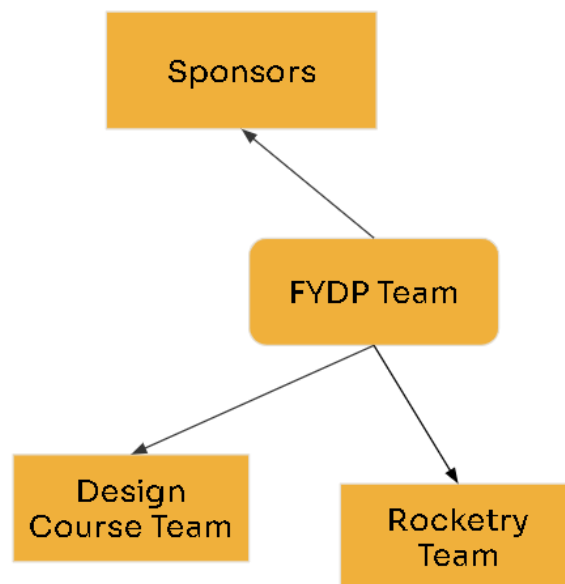


Figure 43: Organizational Chart

5.5 Project Budget

The University provides \$ 75 per student over each term. This funding is expected to run out very quickly. Industry sponsorships through the Waterloo Rocketry team are expected to cover some of this cost, and a private industry sponsor is willing to cover the remainder. The final project costs, and funding sources are outlined below:

Item	Cost(\$CAD)
Metal stock and other raw materials	2500
Fabrication Costs	1000
3D Printed Components (Combustion Chamber)	8000
Valves, fittings, and actuators	1000
Sensors	500
Enclosures & Hardware	2500
Connectors	1700
Power Supply	1000
Networking Equipment	700
Total	18900

5.6 Sources of Funding

Item	Amount(\$CAD)
UW Funding	750
In-Kind Donations	1500
Industry Sponsors	16650

5.7 Change Management

Change management was built off of the existing Waterloo Rocketry management system. Mechanical CAD is stored on Kenesto, a cloud based document management system. This allows different members on the team to work on the CAD files simultaneously. Electrical CAD and software are tracked using Git and Github, which are industry-standard version control systems also used for the team. Other files are stored on the team Google Drive, ensuring easy access to all of the design documents, spreadsheets, PDFs and any media. This will allow the rocketry team to have access to the FYDP team's work and design after the course is over, ensuring continuity of the project. Team member communication was handled through the Waterloo Rocketry Slack to facilitate communication with the team and client. Finally, all meetings were tracked through meeting minutes and deliverables documents in the drive.

5.8 Lessons Learned

Several valuable lessons were learned through the course of this project. While individual lessons learned have been submitted through a separate form, the key team learnings are outlined below.

- Dividing a project up into a not overly-ambitious nominal scope, with distinct (ie non critical to base function) additional work packages is a very effective method to keep a project on

schedule, as the realities of 4th year engineering often mean the actual time available to do capstone work is less than expected. When (not if) the project faces timeline pressure non critical work packages can be removed to maintain schedule. This strategy resulted in Project Sisyphus being the first Rocketry-Associated capstone in recent years to have a fully validated prototype by symposium (which realistically is when work ends as group members then graduate and go their separate ways).

- Always triple check all sensors and remote actuators (valves, ignition coils) before proceeding with an significant test or action (ie firing the engine) to ensure data is obtained.
- Question requirements! Initially, this project was constraining itself to an aluminum chamber and struggling to close the design due to perceived cost limitations, however after discussing this with our sponsor, he had been willing to fund the more expensive but better performing copper chamber, meaning some time was wasted trying to make the aluminum design close.
- Measure twice, cut once, and always have a copy of your drawing printed out in the machine shop.
- Chinese 3D print suppliers seem to be quite reliable and quick based on our experience, and are an order of magnitude cheaper than North American suppliers. Obviously a local supplier is preferred for sustainability, but if money or timelines are an issue companies like UnionFab are a great resource.
- Getting more eyes on a design sooner is always better. This was sometimes challenging with a group of only 5 people who are all busy 4th years, but having other Waterloo Rocketry team members review designs was very helpful and led to some issues being caught early.

6 Conclusions and Recommendations

The Sisyphus engine has been validated to meet the requirements of a testflight vehicle for the Waterloo Rocketry team, while also building some 'institutional knowledge' and providing physical test data for future regenerative cooling and 3D printed designs. Some work still remains before the Sisyphus engine could launch a testflight vehicle, with the main outstanding tasks being:

- Design of a valve to control the flow of pressurant into the tanks. This valve would conceptually be packaged together with a normally open vent valve with a small orifice to allow filling of liquid nitrous oxide.
- Design mass-optimized bulkheads for the tanks that integrate connection points for the upper airframe, longerons in the injector section, and parachute attachment point.
- Size and fabricate a fincan, including potentially rerouting fuel inlet tubes or adding cutouts to allow the tubes to run outside the outer mould line
- Integrate a logging board for all sensor data into the injector section (optional)
- Integrate the existing standard QD valve into the injector section.

Given that this is not a significant amount of work, and that the highest-risk items have been validated, it is recommended that the team make an effort to fly this engine given the hardware exists to save costs on future testflights. Additional details on the notional vehicle design are included in the appendices as a reference.

Additional work which may be of interest for the Waterloo Rocketry team to conduct in future includes:

- Redesign of the injectors to be a 3D printed part, with cooling channels in the fire face and integrated housings for the injector valves.
- Build or obtain a sufficiently sized ground-side propellant storage tank and pressurization system to support longer burns (30s - 1 minute or more), as the key sponsor for this project has indicated a desire to see this in future, and this would allow thermal steady state to be achieved, and therefore improve validation of the modelling done in this project.
- Design a crash structure to sit around the expensive combustion chamber to ensure reusability with harder landings in the case of main parachute failure. It is likely not feasible to ensure zero damage in a ballistic landing but something would be a lot better than nothing even in nominal landings.

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7 Appendices

7.1 Appendix A: Current Specifications

The most up-to date overall engineering design specification is presented below. Additional, more granular requirements for subsystems and the proposed overall testflight vehicle are presented in an appendix.

Requirement	Relation	Value	Units	Verification	Comments
The combustion chamber shall not melt during operation	N/A	N/A	N/A	Test	Static Fire
The engine shall generate a peak thrust of 7 kN	$\geq$	8	kN	Test	Load cell on test stand
All pressurized components shall be designed with $FOS \geq 2$ on burst at MEOP	$\geq$	2	NA	Analysis	
The engine chamber pressure shall be 450 psi	=	450	psi	Test	Static Fire
Propellant supply pressure shall be between 700 and 1000 psi	$\geq, \leq$	700, 1000	psi	Test	Static Fire
Pressure drop across injector shall be nominally 20% of the chamber pressure	=	20	%	Analysis, Secondary by test	Cold flow test, static fire test (indirect measurement)
The engine shall cost no more than \$1000 CAD per launch	$\leq$	1000	\$ (CAD)	Analysis	List consumables

Requirement	Relation	Value	Units	Verification	Comments
The total oxidizer fill (assuming nitrous oxide) should require no more than 1 standard K bottle	$\leq$	1	K-Bottle	Test	Static fire
The engine shall operate at a nominal O:F ratio of 2	=	2	NA	Test	Static Fire
The engine should be re-fireable with very minimal component replacement (less than 1 hour replacement or refurbishment process for entire engine)	$\leq$	1	hr	Demonstrate	Two static fires in same day
The igniter shall fire for at least 5 seconds	$\geq$	5	seconds	Test	
All systems shall be remotely operable from a distance of at least 700 ft	$\geq$	700	ft	Test	Static Fire

Table 9: Engine System Requirements and Verification Methods

Due to the complexity of the integrated system, it has been broken down into the following components: vehicle, engine (combustion chamber and supporting fluid systems) and electrical ground support equipment (EGSE). Note that most of the EGSE is the scope of the Waterloo Rocketry team and already exists, with the key scope of this FYDP team being a comprehensive upgrade of these systems. This section will present the high-level requirements for the vehicle and engine, as well as the engineering design specification. Component-specific requirements will be presented in the sections focused on design progress.

7.1.1 Vehicle

Criteria	Rationale
The vehicle shall be 8 inches in diameter	This is the same size as the competition vehicle and will allow for representative testing
The vehicle shall accommodate payload sections of max length 80in from top of engine to tip of nosecone with a 40in nosecone	This is approximately 20% larger than the existing payload to allow for some design flexibility in future years
The vehicle shall accommodate payload mass up to 20kg (approx. 44lbs)	Current payload weights are about 35 pounds, this will allow for modest weight increases without hindering payload design
The vehicle shall reach a target apogee of 7000 ft AGL	This requirement is set by the waiver provided for the Tennessee launch site
The vehicle shall be statically stable with 2 calibers of stability within wind speeds of 15mph	This is the standard stability as recommended by Launch Canada
The vehicle shall have off-rail velocity higher than 30.5m/s (100ft/s)	Standard practice as recommended by Launch Canada)
The vehicle must include a quick disconnect which can be disconnected reliably just before flight or at takeoff	Need a way to disconnect the tank fill lines in order to fly
The vehicle and all infrastructure required for launch shall be transportable within 1 trailer	This will allow for easy transportation of the vehicle as SDC truck and trailer drivers are hard to find
The vehicle shall be launchable with 4 people as described: 2 ops - 1 with electrical knowledge, 1 with propulsion knowledge; 1 primary operator (trainable); 1 secondary (has done cold flow or static fire test)	Each subsystem testing on any given flight will want 2 people. Limits operator count to 4 due to logistics (only 2 vehicles, 4 seats). Others too busy to train for ops

Constraints	Rationale
All components likely to be damaged in off-nominal situations should be cheap and easy to replace (including fins, nozzle, nosecone, etc.). Ideally, this should be possible in the field as much as possible.	If this vehicle flies more than a few times, it will have off-nominal situations. For it to continue being used by the team, it must be cheap and easy to fix and reflly.

Engine should provide mounting clearance/locations for through-wall fins or other similar fin mounting schemes	Tip-to-tip fins are difficult to manufacture. Engine design can allow for easier fin attachment in the future. Want fins to be replaceable at the launch site, so bolt-on is a big win.
All components likely to require fiddling with at launch should be easy to access. Ideally with the tower raised, second ideally with the rocket on the rail, lastly with the rocket in the trailer.	Likelihood of successful launch increases dramatically with ease of servicing the rocket.
The vehicle should have a combined boost and coast phase flight length of as many seconds as is feasible	Helpful to gather useful data for controls projects.
The vehicle shall follow the latest Launch Canada DTEG and rules and regulations	Easy way to prove safety best practices are being followed and get approval for launch/testing.

7.1.2 Engine

Criteria	Rationale
The engine shall generate a minimum thrust equal to 10 times the entire vehicle mass	The rocket is required to reach 100 ft/s before leaving the rail, requiring a high TWR. (10–12x is Launch Canada recommendation)
The engine shall cost no more than \$1000 CAD per launch (not including case etc.)	This will ensure that test flight is economically viable in the future
The engine shall use Nitrous Oxide and Ethanol as propellants	These propellants are easy to source both in Canada and the US. The team has prior experience with them
Engine shall have hard mounting locations at bottom and top to attach to the rest of the vehicle	The engine must easily integrate
Propellant tanks shall be 8 inches in diameter	Matching the vehicle-level requirement to avoid the need of fairings
The propellants shall be supercharged with nitrogen	Compared to previous WR designs which used ambient heating, this will reduce testing times by up to 1.5 hours in winter conditions, eliminate the need for tank heating solutions, and allows for denser nitrous to be used which improves efficiency.]

The engine shall be designed to interface with existing test stand	To reduce the amount of work the FYDP has to do by using existing team parts. This includes combustion chamber size (max 6" OD) and compatibility with the beam load cell for thrust measurement
All pressurized components shall be designed with $FOS \geq 2$ on burst at MEOP	Safety guideline. As per Launch Canada guidelines
Any pressurant systems shall have a MEOP < 2500 psia	Conservative value based on full N ₂ bottle pressure. Allows filling without a boost pump
The propellant tanks shall have a MEOP of 1000 psia	Give significant margin for pressure drops across regen cooling and injectors, accounting for blowdown, while maintaining performance. Also a relatively standard PRV setpoint.
The propellant tanks shall be sized such that tank pressure is > 650 psia at engine shutdown (liquid depletion)	Ensures adequate pressure drop margin over regen channels and injectors for the entirety of the burn.
The pressure drop across the injectors shall be at least 20% of the chamber pressure	From NASA SP-8089. Minimum to avoid chugging and coupled feed system instabilities
The combustion chamber shall operate at a MEOP of 400 psia	Based on previous WR liquids. Higher chamber pressure yields more thrust for the same massflow but reduces margin for tank pressure drop
The total oxidizer fill (assuming nitrous oxide) shall be no more than 12 kg	Fits within a single K-type cylinder. Legal transport limit is 2 cylinders/vehicle. Enables 1 vehicle to carry supplies for 1 launch attempt. More scrubs likely require returning to Waterloo

Constraints	Rationale
The engine burn time should be at least 3 seconds	This should allow the engine to reach a thermal steady state if regenerative cooling is used
The engine should be re-fireable with no component replacement	This could allow for multiple (e.g., 2) flights in a single sortie. Especially beneficial for controls, who may need to tune and re-test controllers before flying on a competition vehicle
The engine should be pressure-fed by a regulated or otherwise controlled supply of gaseous nitrogen	This allows testing and launch operations to be faster and independent of heating methods or weather conditions

The engine should be regeneratively cooled	A challenging and interesting design that also significantly extends engine life. Enables extended burns with stretched tanks if lower TWR is permitted (e.g., with active control during boost)
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7.1.3 Electrical Functional

Constraints	Rationale
Shall support up to 10 motorized ball valves or igniters in any combination	Based on existing ground side plumbing and pyrotechnic igniter design, which is not expected to change significantly
Shall support data acquisition from at least 12 4-wire sensors	Based on project and team sensor requirements
Shall provide at least 2 parallel interfaces to the ground side CAN network supporting at least 4 pins, 2A current each	Necessary to interface with the custom sensor PCB at tests, and with the rocket during launch
Shall be remotely operable from at least 1000 ft away	Mission control must be beyond the calculated standoff distance
All connectors carrying over 25V shall be of appropriate polarity given direction of power flow	Risk of shorting and personnel contact with live power; 25V is below most standards (30-60V) presented in [11]
No SRAD electronics shall directly interface with 120VAC input power	Presents a significant personnel hazard, see [11]
Actuator power shall be controlled with a dedicated lockout key switch	Launch Canada DTEG requirement [12]

Criteria	Rationale
System should be water and debris-proof (meet IPx6 or NEMA 3x/4x/6x ratings as applicable)	Want to be able to test and launch outdoors in all weather; Testing is often conducted with high pressure water; Metal hoses pose a known shorting risk
AC power (generator) failure should not de-power the system	Generators have been known to fail sporadically, forcibly aborting tests or launch. System should be able to run on generator power when available and fail over to another source
All connectors for a given system (sensors, actuators, power feed) should be unique	Prevents incorrect connections from being made to/from the system with an otherwise properly wired cable (e.g. 24V tank heating power into a sensor input port)

Battery backup should be passively recharged from 120VAC	Limits need to take apart system and recharge car batteries
All electrical ground support equipment should be assembled in the field by non-electrical personnel in 30 min or less	Limited personnel at a test flight launch, system experts may not be present
All electrical ground support equipment should transportable safely by one person	Do not want to be reliant on multiple people present

7.2 Appendix B: Risk Register

At this time, all technical risks have been retired given successful testing has occurred. This risk register is a relic of the risks foreseen early in the project, the mitigations that were implemented.

Risk Type	Risk	Severity	Likelihood	Mitigation	Rating 1- Insignificant 2- Somewhat important 3- Critical
Schedule	3D printing lead time for engine too long or print fails	High	Medium	Chose suppliers based on low lead times. Designed to not push the limits of machine capabilities. Print did not fail.	3
Schedule	Insufficient time/man-power to machine and fabricate all components.	High	Medium	Regular reviews of schedule. Keep scope reasonable by re-using existing team designs that will work with our vehicle architecture. Outsourced where possible.	1
Technical	DAQ or Logging PCBs do not work	Medium	Medium	Existing team systems could be used, albeit with significantly more work and headache - new systems are intended to be an upgrade. New systems ended up working.	1
Technical	Thermal design insufficient, chamber melts	Medium	Medium	As much safety factor as possible on heating. Printed representative test article to verify design parameters. Chamber did not melt.	3

Technical	Flow distribution around regen cooling channels uneven	High	Moderate	Dedicated CFD on fuel distribution volute and testing on final part showed acceptably uniform flow. Changing OF ratio to lower temperature and increase fuel flow rate also considered as an option.	3
Technical	Valves freeze	High	Low	Use previously tested valves. Always have multiple ways to vent tank. Freezing occurred during one test, backup valves successfully safed system.	2
Technical	Leaks in engine system	Medium	Medium	Used standard practices for fitting use. Used assembly procedures.	2
Technical	Engine mass too heavy for vehicle	High	Low	Reviewed mass regularly, tracked / predicted in a mass tracking doc	2
Technical	Engine complexity too high for use as test flight vehicle	Medium	Medium	Designed from the start considering the user. Reused team architecture choices.	2
Technical	Igniter does not work	High	Very low	Used teams previously tested formula for RNX, tested early to ensure ignition	1
Schedule	Inability to secure funding from main sponsor	High	Low	Secured buy in to project early on from external sponsor	2
Schedule	No access to suitable test facility.	High	Low	Access to current James Downey Way site confirmed provided all by-laws are obeyed.	2
Technical	Temperature sensor response time insufficient	Medium	Medium	Thermocouples obtained and tested, response time slower than desired but acceptable.	2

7.3 Appendix C: Work Breakdown Structure

Table 16: PROJECT: SYSPHUS Hours Breakdown - Part 1

Task	Hours
Project Definition and Research	50
Problem Definition	5
Consulting With Stakeholders	5
Requirements & Specifications	10
Project Management Definition	10
Research (Overall)	20
Design	868
Combustion Chamber	110
Design Volute to Attach to end of Combustion Chamber	30
Validate Volute CFD Results with Testing	15
3D Print Test Article	15
Develop Model to Correlate Test Data to Actual CC	10
Finalize Combustion Chamber CAD	10
3D Print the Chamber	15
Pressure drop testing etc.	10
Integration into rest of system	5
Fill out COA application with Peter	3
Fill out forms etc	3
Igniter	30
Initial Research / Testing Plan	5
Preliminary Design	5
Testing	10
Final Design	10

Table 17: PROJECT: SYSPHUS Hours Breakdown - Part 2

Task	Hours
Pressurization/feed System	130
Architecture Choice	10
Initial Sizing	10
Component Sizing / Selection	10
Design	30
Machining/tube bending/fabrication	30
Electrical design and fabrication and testing (Maybe not needed)	20
Integrated Testing	20
Electrical System	90
Electrical Design	60
Electrical testing	30
Injectors	160
Initial Design	20
Prelim Testing	20
Design	60
Full Scale Testing	60
Valves	70
Sizing and drawing creation	30
Machining	20
Testing	20
Disconnect system	55
Design	20
Drawing creation	5
Manufacturing	10
Testing	20
DAQ PCB	110
Requirement Definition	5
Schematic Creation	15
Design Calculations	10
Simulation	10
PCB Layout	10
Review	5
Dev Board Fabrication	15
Write Firmware	15
Dev Board Characterization and Testing	5
Final Board Design	10
Final Board Fabrication	10

Table 18: PROJECT: SISYPHUS Hours Breakdown - Part 3

Task	Hours
Data Logging PCB	70
Requirement Definition	5
Schematic Creation	10
PCB Layout	10
Review	5
Dev Board Fabrication	5
Firmware	15
Dev Board Testing	5
Flight Board Design	10
Flight Board Fabrication	5
Fluids GSE	40
Component Selection and Sizing	10
Frame Design and System CAD	10
Component Ordering and Fabrication	5
System Assembly	15
Testing and Validation	400
Static Fire	370
Manufacture of Components	50
Integration with GSE	20
Cold Flows	200
Static Fire	100
Catch up on Documentation	30
Write payload user guide	20
Clean up detailed design documentation	10
Deliverables	220
Design Proposal Report	10
Design Proposal Presentation	10
Midterm Presentation	10
Final Report	15
Final Presentation	15
Meetings With Dr. Milne	160
Total	1538
Total + Margin (5%)	1614.9

7.4 Appendix D: Gantt Chart

A Gantt chart has been created in google drive which ensures an up to date copy can be easily accessed or edited by all group members at any time. A screenshot of this file is shown in Figure 44. It contains a lot of information which is hard to read so it is easier to read by accessing digitally here:

Gantt Chart

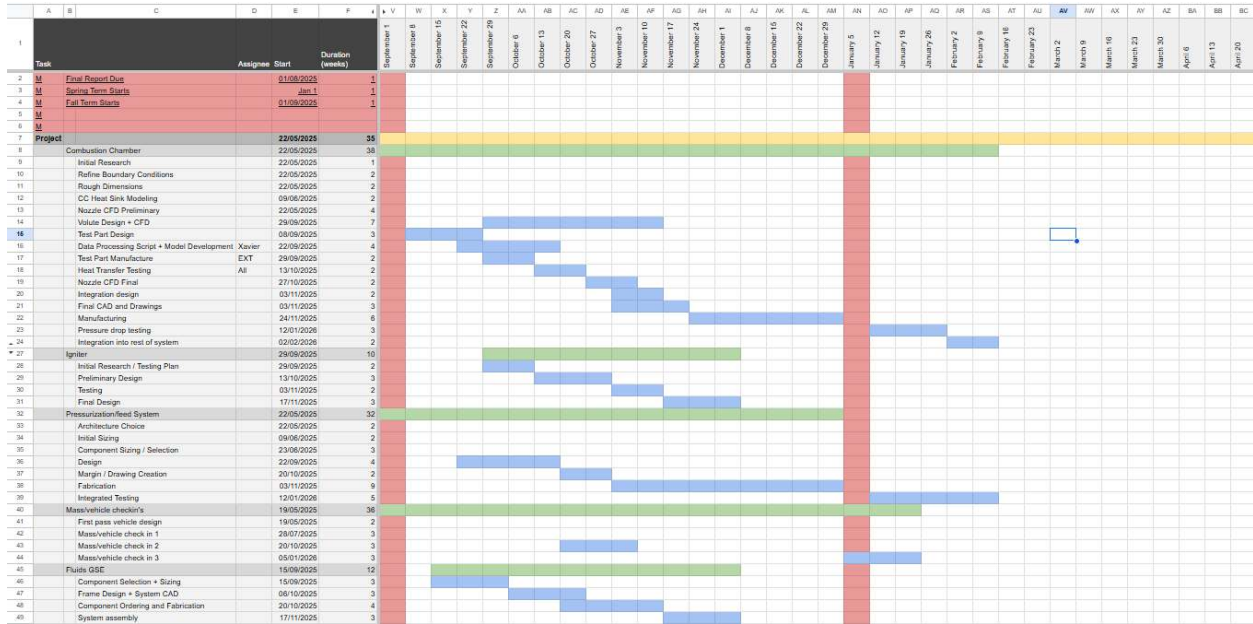


Figure 44: Gantt Chart

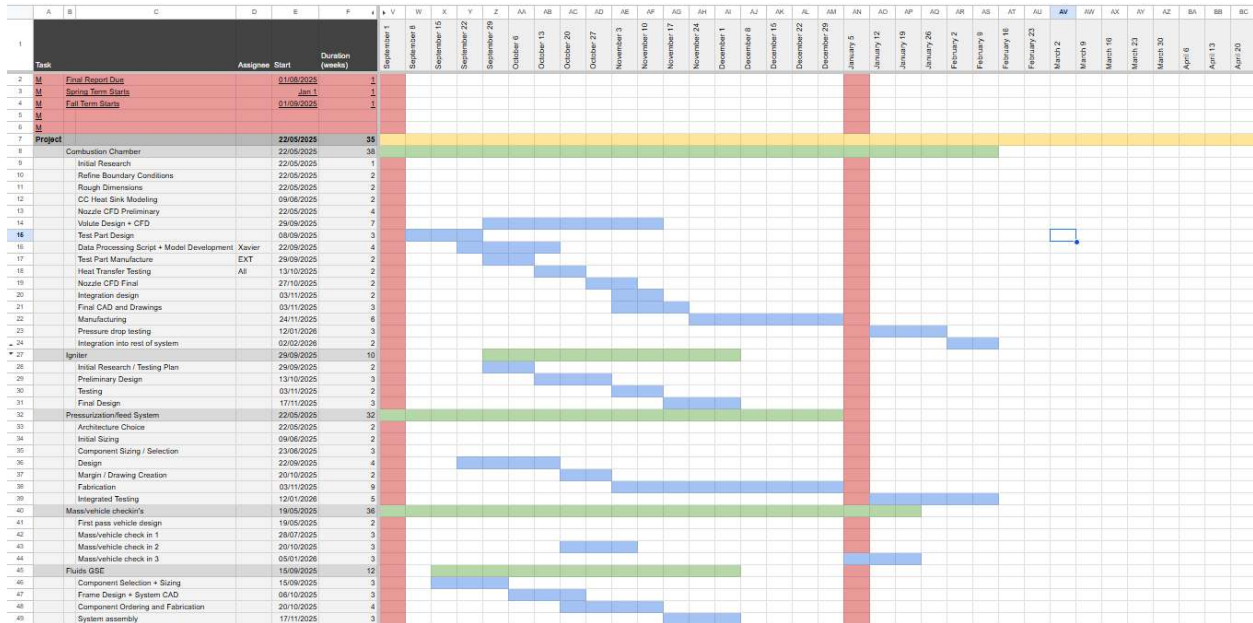


Figure 45: Gantt Chart

7.5 Appendix E: Test Flight Vehicle Design

Because Sisyphus was designed to fly with a purpose, in order to design the engine some level of vehicle design was required to set engine requirements. This was done using previous team experience in competition vehicle design and test flight vehicle design.

The test flight vehicle was assumed to have a payload (everything except the engine) of 50 pounds. An apogee of 7000 feet. A launch rail that is 20ft long (longest easily acquirable at the test site according to team advisor). And a desired off rail velocity over 100 feet per second. In order to determine engine requirements an OpenRocket simulation was done assuming early Sisyphus design mass estimates. This resulted in choosing a thrust of 7kN and using a 4 second burn time with thrust roughly linearly decreasing to 3.5kN at the end of the burn (propellant tank blow down). The engine thrust level assumed is shown below in Figure 46. This did end up being fairly similar to actual performance as well.

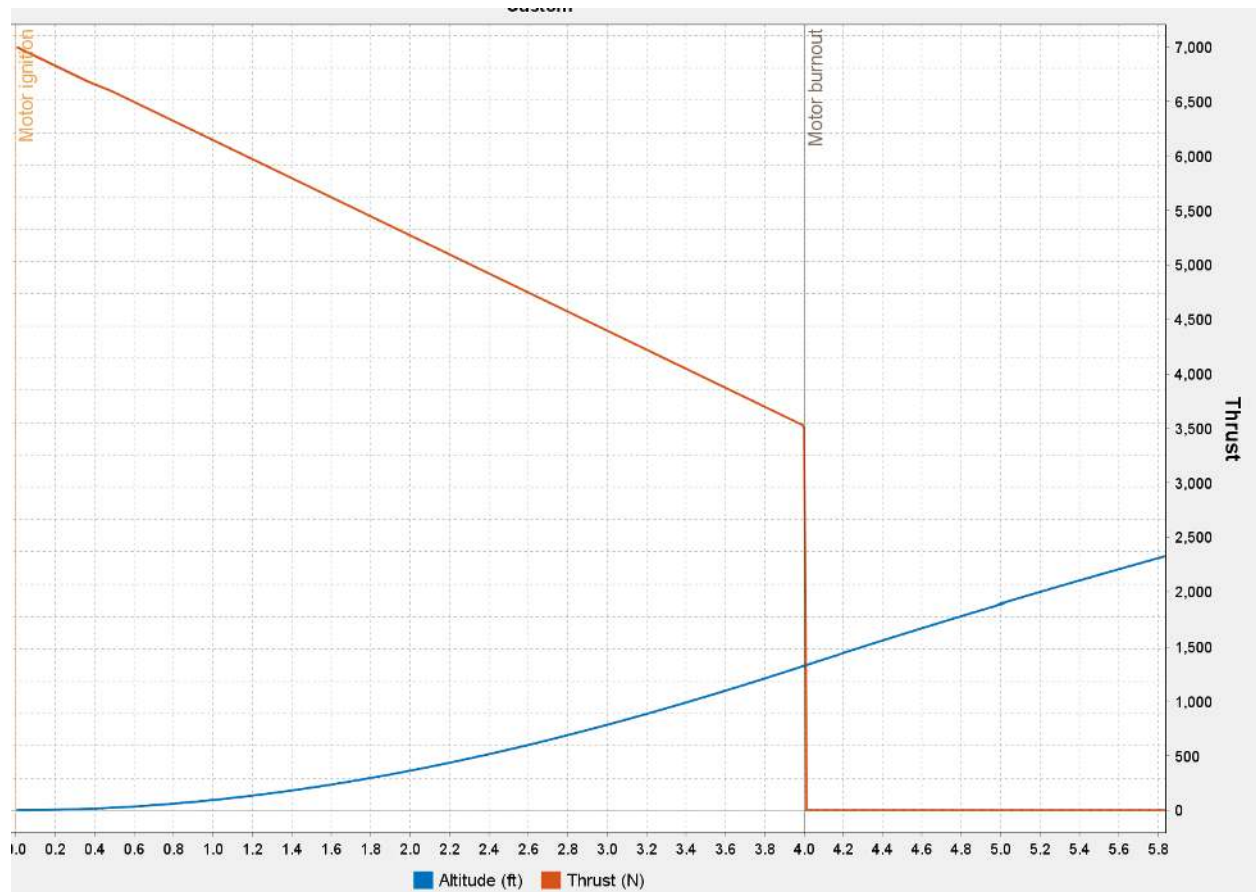


Figure 46: Vehicle Design Assumed Engine Performance

In reality, for future use of the engine it is recommended to start with updating the Sisyphus engine mass by weighing the major component since an as fabricated engine mass was not calculated or measured. Next create a new OpenRocket file with desired payload mass, and vehicle geometry

(length, fin shape etc.). Once this is done, start with the RSE file showing 7kN at the start and linearly decreasing to 3.5kN over 4 seconds. This will be a good starting point. Note that the tanks can support a slightly longer burn than this and it is highly likely the tank created in this project will be sufficient to overshoot the altitude waiver at the test site. If this is the case then adjust the length of the engine burn until desired vehicle performance is met. It can be assumed thrust will still linearly decrease from 7kN to 3.5kN with a shorter burn time due to pressure blowdown in the tanks and a propulsion team member can easily calculate how full to fill the tanks for various burn lengths. If on the other hand the existing tanks are not sufficient or much higher altitudes and speeds are desired it could be very possible to fabricate a new tank and fly much higher and faster. It is estimated from previous experience that this engine could fly to around 30 000ft without an incredible amount of difficulty depending on the payload mass and tank length increase used. Openrocket and RSE files can be found in the Waterloo Rocketry team google drive. Additionally ChatGPT can be utilized to generate rapid turn around RSE files with any simple thrust characteristics.

7.6 Appendix F: Calculations, Test Data, and Procedures

Calculations and raw data are stored on the Waterloo Rocketry team's Google Drive and is publicly viewable. Some key spreadsheets used are linked below. [Final Design Review Presentation with line-by-line verification](#)

[Static Fire Test Data](#)

[Coldflow Test Data](#)

[Static Fire Operations Procedure](#)

[Tank Pressure Test Data Analysis](#)

[Tank Design Spreadsheet](#)

[Injector Design Spreadsheet](#)

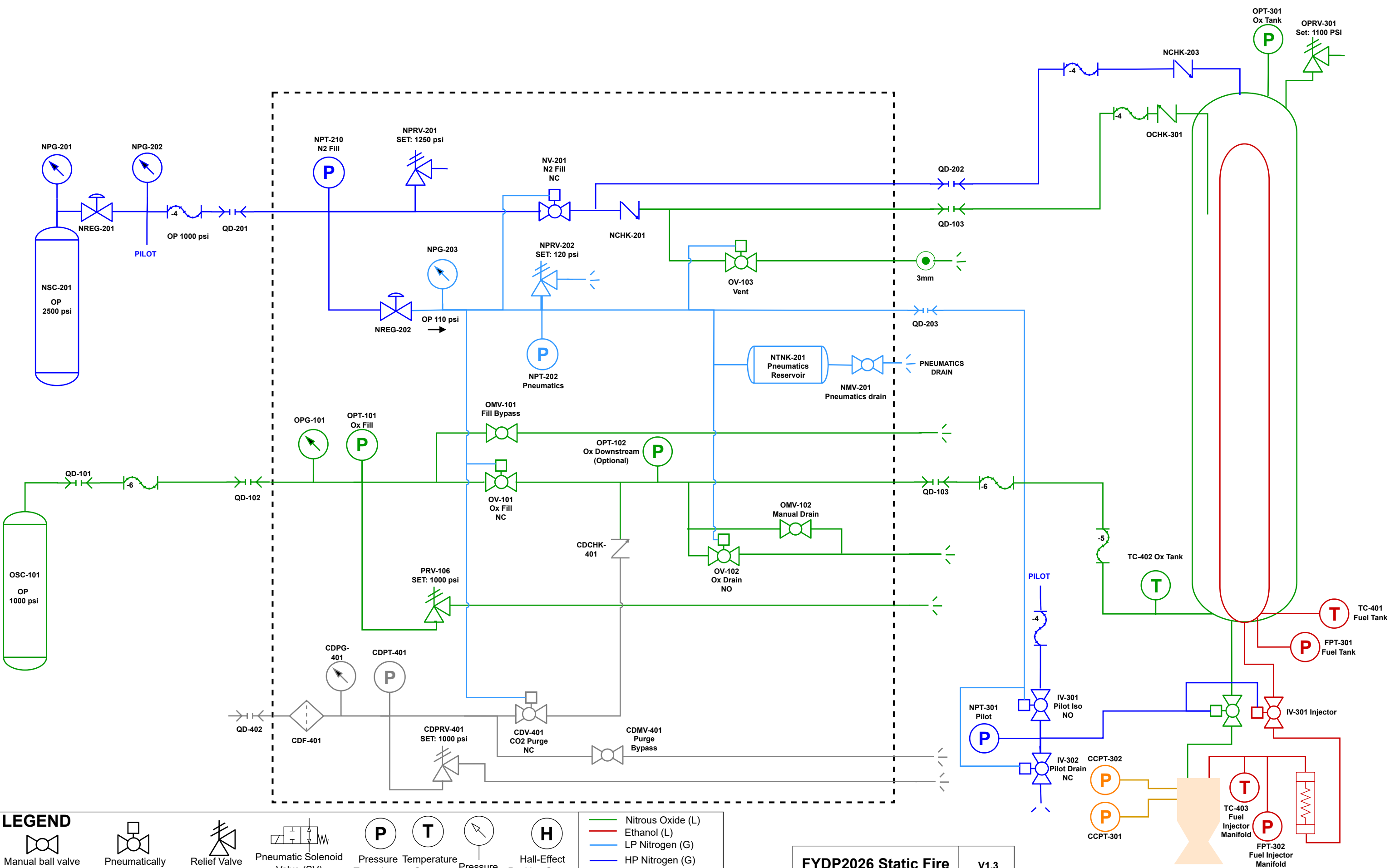
[DAQ System Design Log](#)

[Injector Testing Data](#)

[Static Fire Video Footage](#)

7.7 Appendix G: P&IDs

The P&ID engine and fill cart in test configuration is included on the following page.

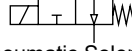


LEGEND


Manual ball valve (MV)


Pneumatically Operated ball valve (V)


Relief Valve (PRV)


Pneumatic Solenoid Valve (SV)


Pressure Transducer


Temperature Sensor


Pressure Gauge


Hall-Effect Position Sensor


Remote valve (V)


Regulator (REG)


Check Valve (CHK)


Filter (F)


Orifice


Vent


Hose


Quick Disconnect (QD)

Nitrous Oxide (L)

Ethanol (L)

LP Nitrogen (G)

HP Nitrogen (G)

Carbon Dioxide (L)

NO = Normally Open

NC = Normally Closed

OP = Operating Pressure

FYDP2026 Static Fire	V1.3
Last Revised by: Matthew Gordon	
2026-03-12	

7.8 Appendix H: Drawing Package

In order to limit page count of this document, all drawings have been uploaded to a publicly-viewable google drive folder.

[Drawing Package](#)

7.9 Appendix I: Hazard Disclosure Form

The Hazard Disclosure form for Team 15 is included on the following pages.

ME481 F25 Hazard Disclosure Form

Team Number: 15

Project Title: Project Sisyphus: Regeneratively Cooled Rocket Engines for Student Rocketry

Brief Project Description (2-3 sentences):

This project aims to develop a regeneratively-cooled liquid rocket engine for use by the Waterloo Rocketry student design team as part of a test-flight vehicle. The engine will use Nitrous Oxide and Ethanol as propellants.

Answer all the following questions. If unsure, answer yes. **If you answer yes to any question you need to attach your own page which provides a concise and quantitative description of the items, use, and safety measures for each item.**

Hazard Classification	Hazard exists during:		
	Fabrication / Manufacturing	Symposium Demonstration	Usage - Post Symposium
1. Is electricity used for anything other than stand-alone unmodified computers?	yes ☒	yes ☐	yes ☒
2. Are any lasers used? See UW Laser Program .	yes ☐	yes ☐	yes ☐
3. Are any flashing or strobe lights used?	yes ☐	yes ☐	yes ☐
4. Are any lights brighter than a 60W bulb or 800 lumens?	yes ☐	yes ☐	yes ☐
5. Are there any radiation sources? See UW Radiation Safety .	yes ☐	yes ☐	yes ☐
6. Will there be any X-Rays emitted? See UW X-Ray Program .	yes ☐	yes ☐	yes ☐
7. Is there any combustion occurring?	yes ☐	yes ☐	yes ☒
8. Are any temperatures created above 45°C or below 10°C ¹	yes ☐	yes ☐	yes ☒
9. Are any compressed gasses used? See UW Compressed Gases .	yes ☐	yes ☐	yes ☒
10. Is there any significant stored energy (electrical, chemical, mechanical) in any component?	yes ☐	yes ☐	yes ☒
11. Are any chemicals used other than compressed gases described above? This includes gases, liquids, powders, and solids. See UW WHMIS . Attach MSDS sheets.	yes ☒	yes ☐	yes ☒
12. Are any nano-sized objects used or made? This would include materials that have external dimension less than 100nm.	yes ☐	yes ☐	yes ☐
13. Are there any biological components? This includes (both dead and alive): blood, tissue, fluids, parts, from any organism from bacteria to live humans, or live test subjects. See UW Bio-Safety .	yes ☐	yes ☐	yes ☐
14. Is any food for human or animal consumption used?	yes ☐	yes ☐	yes ☐

¹ <https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20100020960.pdf>, accessed 190106

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15. Are any gases, particles, or fluids being ejected into the environment?	yes ☐	yes ☐	yes ☒
16. Are there any moving parts?	yes ☐	yes ☐	yes ☒
17. Is any component or the entire project heavier than 10kg ?	yes ☒	yes ☒	yes ☒
18. Can any part fly?	yes ☐	yes ☐	yes ☒
19. Can there be any projectiles?	yes ☒	yes ☐	yes ☒
20. Is any noise generated above normal speaking voice levels?	yes ☐	yes ☐	yes ☒
21. Do you know of or suspect there might be any other hazards?	yes ☐	yes ☐	yes ☐

Please **write a concise and quantitative statement** to describe each potential hazard and identify your proposed mitigation procedure for each item. Note that you should consider the following hierarchy of Hazard Control:

- Eliminate the hazard
- Substitute a less hazardous component or process
- Implement engineering controls to minimize the hazard
- Implement administrative controls to minimize the hazard
- Use appropriate Personal Protective Equipment (PPE)

"The most effective way to deal with a hazard is to remove it"

Hazard Classification #1

Electricity is used in the testing and operation of the electrical support equipment, which includes custom circuit boards and field wiring. Power is supplied from lipo batteries, car batteries, or a generator (for the ground support systems during testing), meaning no custom electronics will be plugged directly into the power grid. If the appropriate CSA certification can be acquired, then the system may also be used with standard wall outlets. All components are outfitted with fuses and/or breakers to prevent overcurrent events. The system is designed to be waterproof and resistant to vibration, debris, etc. The system will not be powered at symposium, again unless CSA approved.

Hazard Classification #2:

Combustion will occur when the engine is tested. Combustion can only occur when propellants are loaded into the system, which will only occur under highly controlled circumstances. The system will not be filled with any propellants during symposium, so combustion cannot occur. During propellant loading and whenever personnel are near the system physical lockouts (a disarming key) will be installed to prevent accidental combustion, and the wires connecting the igniter will be physically unplugged. Combustion will only occur when operators have retreated to a safe distance. See the attached document for more information on this safe distance. All testing will be conducted in an area that will be closed off to the public to eliminate the hazard to passersby.

Hazard Classification #3:

Temperatures below -10°C can be expected during propellant fill, particularly when opening cylinders of compressed gas, and temperatures well above 45°C can be expected during hotfire testing. As with combustion, this cannot occur unless propellants are loaded. The same mitigations from hazard #2 apply, with physical lockouts preventing propellant fill into the system until operators are clear of the area. As an additional precaution, all operators at the test site during setup (opening the compressed gas cylinders) will wear clothing which fully covers their bodies, and a pair of cryogenic-rated gloves.

Hazard Classification #3:

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Compressed gases including nitrous oxide, nitrogen, and carbon dioxide will be used. The same mitigations as detailed in hazards 2 and 3 apply. All members of the team have completed the UW compressed gas safety training and will follow all required handling procedures.

Hazard Classification #4:

Significant chemical energy is stored when the system is filled with propellant. Mitigations from above hazards apply and ensure that the system only contains this energy when operators have retreated to a safe distance – before this, the energy is stored in standard compressed gas bottles, which will be handled according to the UW compressed gas training.

Hazard Classification #5:

Potassium Nitrate and Epoxy are intended to be used in the igniter system. Potassium Nitrate is an oxidizing powder which can also irritate the eyes and respiratory system. Manufacturing of this substance will be carried out in a well-ventilated location, open/stored volumes will be limited as much as possible, a category D fire extinguisher will be present in the location where Potassium Nitrate is used. Team members will wear full PPE (including a properly fitted respirator) and a team member will be posted at the entrance to ensure no external persons enter the space while manufacturing is occurring.

For epoxy resin and hardener, these substances can irritate skin and can also begin a runaway exothermic reaction. To mitigate this, volumes will be minimized, no exposed skin will be permitted, manufacturing will occur in a well-ventilated space, and full PPE will be worn as above.

Hazard Classification #6:

During testing, nitrous oxide, ethanol, and carbon dioxide may be released to the environment in the event of an abort, and combustion products will be released during a successful test. The ground support plumbing shall be designed such that only nitrous oxide is released automatically if an aborted test occurs, both reducing the combustion risk and allowing ethanol to be manually drained and re-used/disposed instead of being released. Both nitrous oxide and carbon dioxide are naturally-occurring substances which will release as gases. Combustion products will be made up largely of water, carbon dioxide, and some nitrous oxides, all of which are either naturally occurring or are present in common systems such as automobile exhausts. Release of gases will only occur within a closed-off area free of people, and in the event of a release personnel will not be allowed to re-enter the site until the vapor cloud is no longer visible indicating the gases have safely dispersed.

Hazard Classification #7:

Moving parts exist, including valves and actuators. These components will all be internal, meaning that extremities cannot become trapped or jammed in a moving part. No motion is planned for the symposium, as the moving parts are not visible to an external observer (and most require pressurized gases to create motion).

Hazard Classification #8:

The entire system is expected to be heavier than 10kg. Items such as ground support equipment, which will be the heaviest components, will be designed with carrying handles or wheels to allow multiple people to split the load. The fully assembled system will always be transported by multiple people to ensure no injury. Additionally, the system will not be filled with propellant during transportation, reducing the mass that must be lifted.

Hazard Classification #9:

The rocket engine is designed to be installed on a vehicle that will fly. During hotfire testing, the engine will be rigidly mounted to prevent flight. Before flying on a vehicle (which is outside the scope of this FYDP) an airspace closure will be obtained from the proper authorities (FAA/Transport Canada), and the vehicle will be launched at an angle away from any personnel.

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Hazard Classification #10:

Projectiles may occur during fabrication (machining) and in the event of an explosion or test anomaly. During machining, proper practices will be used to prevent large amounts of chips from being ejected (monitoring cutting speeds and depths), and proper PPE will be worn as required by ESMS regulations. Manufacturing which may generate chips or debris will occur in an area such as the ESMS where all people present are trained and wearing protective equipment.

In terms of debris generated from an explosive event or test failure, the clear zone that will be enforced during testing has been sized based on FAA standards and using conservative assumptions to limit both the overpressure wave magnitude and the fragment density. See the attached document for the full calculations and methodology.

Hazard Classification #11

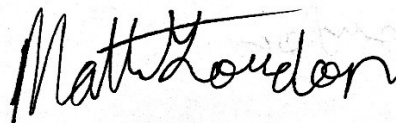
Noise will be generated during engine testing which is in excess of normal speaking levels. The clear zone around the engine will serve as a buffer to dampen this noise. As additional mitigation, hearing protection will be worn by all team members, with additional hearing protection on hand with perimeter guards around the clear area to offer to any passersby who are near the zone at the time of the test. Additionally, signage will be posted in advance of the test warning about the hazards and noise, and testing will be conducted within the hours allowed for by Waterloo By-Laws.

Faculty Advisor: Dr. Andrew Milne

Date disclosure completed: September 18th 2025

Completed by: Matthew Gordon

Signature:

A handwritten signature in black ink that reads "Matthew Gordon". The signature is written in a cursive, flowing style with a large initial 'M'.

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APPENDIX A: BLAST ZONE CALCULATIONS

The safety calculations in this document are based on U.S. Federal Aviation Administration (FAA) guide 437.53-1, Calculation of Safety Clear Zones for Experimental Permits under 14 CFR § 437.53(a) (U.S. Department of Transportation) [1] and U.S. DoD Defense Explosives Safety Regulations 6055.09 Edition 1, Change 1 [2].

Calculation methodologies are copied from previous Waterloo Rocketry team reports, notably the report on the Beringer Field Blast Zones [3].

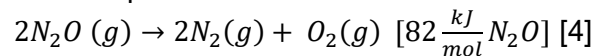
The Sisyphus engine uses Nitrous Oxide as an oxidizer and Ethanol as a fuel, as with previous team liquid engines.

The Sisyphus engine is designed to operate at an oxidizer to fuel (O/F) ratio of 2, which is fuel-rich, and therefore oxidizer is the limiting reagent. The total mass flow rate of the engine is 3.75 kg/s, with a maximum burn time of 4s during flight. A longer duration, 10s burn is also under consideration for testing to validate the effectiveness of the regen cooling.

As a conservative assumption for all calculations below, the total propellant mass is re-distributed to be in stoichiometric proportions - this is the highest-energy combination for a given total propellant mass, which results in the most conservative blast distance.

Nitrous Oxide

Nitrous oxide can undergo thermal decomposition at standard conditions according to the equation



Converting this to an explosive yield factor, which represents the equivalent mass of TNT to match the energy stored in an explosive, with 1g of TNT having an energy of 4.184 kJ [5].

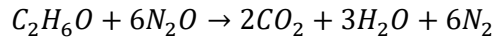
$$Y_{Nitrous} = \frac{\frac{82 \frac{kJ}{mol}}{44.0128 \frac{g}{mol}}}{4.184 \frac{kJ}{g}} = 0.445$$

Nitrous Oxide and Ethanol

Compound	Molar Mass (g/mol)	Heat of Combustion (kJ/mol)
Nitrous oxide	44.0128	N/A
Ethanol	46.0684	1367.6 [6]

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The overall reaction for complete ethanol combustion with nitrous oxide is as follows:



Note that the energy of combustion for ethanol is for the combustion of ethanol with diatomic oxygen. The energy of decomposition of nitrous oxide is omitted from this equation as it is calculated using the yield factor found previously.

The molar stoichiometric ratio between oxidizer and fuel is observed to be 6. Adjusting for molar masses, the mass based stoichiometric ratio is 5.732. This means for every 1 kg of ethanol loaded, 5.732 kg of nitrous oxide must be loaded for a stoichiometric mixture. A stoichiometric mixture represents the highest possible energy for a given total mass of propellant, so this is a conservative assumption.

Given that the reaction will fully exhaust all propellants in a stoichiometric mixture, the explosive yield of ethanol is:

$$Y_{Ethanol} = \frac{\frac{1367.6 \frac{kJ}{mol}}{46.0684 \frac{g}{mol}}}{4.184 \frac{kJ}{g}} = 7.412$$

The overall yield factor for stoichiometric mixture of ethanol and nitrous oxide is therefore:

$$Y_{Total} = Y_{Ethanol} * \frac{1}{1 + 5.732} + Y_{Nitrous} * \frac{5.732}{1 + 5.732} = 1.481$$

For a 4s burn, the total propellant consumption is $4 * 3.75 = 15 \text{ kg}$. The net explosive weight of this propellant would be:

$$W_{4s} = m_{prop} * Y_{Total} = 22.215 \text{ kg} = 48.98 \text{ lbs}$$

For a 10s burn, the total propellant consumption is $10 * 3.75 = 37.5 \text{ kg}$. The net explosive weight of this propellant would be:

$$W_{10} = m_{prop} * Y_{Total} = 55.54 \text{ kg} = 122.445 \text{ lbs}$$

According to (U.S. Department of Transportation), an acceptable safe limit for blast overpressure is 1.0 psi, so this analysis will use this figure. According to the Kingery-Bulmash relationship, for an overpressure of 1.0 psi, the factor relating net explosive weight to distance, K, is $45 \text{ ft} \cdot \text{lb}^{1/3}$. Given the maximum net explosive weights above, and the following equation, the minimum safe distances for blast overpressure can be computed:

$$D = KW^{\frac{1}{3}}$$
$$D_{4s} = 164.65 \text{ ft}$$
$$D_{10} = 223.46 \text{ ft}$$

The FAA explosion safety document suggests a maximum hazardous fragment density of 1 fragment per 600 ft², with each fragment having an impact energy of 79 J or less, as an acceptable safe limit. An equation from

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the Department of Defense Manual 6055.09-M-V3 [2] is used to calculate a minimum safe distance assuming no shielding or protection, called the hazardous fragment distance (HFD).

For $W < 100$ lb:

$$HFD = 291.3 + [79.2 \ln(W)]$$

For $W \geq 100$ lb:

$$HFD = -1133.9 + [389 * \ln(W)]$$

From these equations:

$$HFD_{4s} = 599.5 \text{ ft}$$

$$HFD_{10} = 703 \text{ ft}$$

The HFD values are larger than the overpressure wave values, so these will be used to size the blast zone.

Figure 1 shows the hazardous fragment distances overlayed on Google Maps centered at the Waterloo Rocketry Team's current testing location.



Figure 1: Blast Distances Overlayed on Google Maps

Note that the only overlap with a privately-owned space is the southeastern corner of the Stork Family YMCA parking lot, which is covered only by the HFD for the 10 second burn. Efforts will be made to keep people and vehicles out of this section of the parking lot, however if this is not possible (if passersby do not choose to comply with requests) the following additional actions are also taken by default:

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- The nozzle of the engine shall be pointed in away from the parking lot, placing the team's shipping container between the engine parking lot and adding impedance to fragments travelling in that direction. The container is equipped with sandbag-filled walls which should entirely stop fragments in this direction, however since the team does not have any expertise on blast shielding available the effect of this shielding is entirely ignored in clear zone calculations.
- The actual propellant combination will be fuel-rich of stoichiometric, meaning the actual energy present will be less than the value used in calculations of the HFD (due to unburnt residual propellant which will not contribute to an explosion).
- The calculations above assume 100% mixing and complete detonation of all propellants. Given the propellants will be stored in separate tanks, this is a relatively conservative assumption, once again reducing the 'real' HFD zone diameter.
- Passersby will not be permitted to leave the parking lot, as the edge of the lot marks the start of University of Waterloo owned private land which can be closed with proper permission from the Safety Office and Campus Police.

The UW safety office and Campus Police, as well as other department stakeholders, will be informed of proposed testing, and no testing will be conducted before authorization from these authorities is received, following the currently established procedures from the Waterloo Rocketry student design team.

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[OBJ]

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[OBJ]

References

- [1] U.S. Department of Transportation, Federal Aviation Administration, "Calculation of Safety Clear Zones for Experimental Permits under 14 CFR § 437.53(a)," 28 February 2011. [Online]. Available: https://www.faa.gov/about/office_org/headquarters_offices/ast/regulations/media/Guide-Cal-of-Safety-Clear-Zones.pdf.
- [2] U.S. Department of Defence, "Defense Explosives Safety Regulation 6055.09 Edition 1, Change 1," 23 February 2024. [Online]. Available: <https://www.denix.osd.mil/ddes/denix-files/sites/32/2024/03/DESR-6055.09-Edition1-Change-1-240227-Final.pdf>.
- [3] T. P. T. Fairhead, "Engine Testing at Bearinger Field - Suitability Analysis," [Online]. Available: https://docs.google.com/document/d/10tbwypTffey1Lq6TsyNs30xVkfRasQ1qL_-i44l1B0c/edit?tab=t.0#heading=h.puv4zpuzriy3. [Accessed 20 September 2025].
- [4] M. S. V. G. T. L. V. Zakirov, "Surrey Research on Nitrous Oxide Catalytic Decompositions for Space Applications," [Online]. Available: [https://digitalcommons.usu.edu/cgi/viewcontent.cgi?article=2109&context=smallsat#:~:text=In%20the%20past%20nitrous%20oxide,is%20achieved%20\(Rateheat%20generated%20%E2%80%93%20Rateheat](https://digitalcommons.usu.edu/cgi/viewcontent.cgi?article=2109&context=smallsat#:~:text=In%20the%20past%20nitrous%20oxide,is%20achieved%20(Rateheat%20generated%20%E2%80%93%20Rateheat). [Accessed 20 September 2025].
- [5] NIST, "NIST Guide to the SI, Appendix B.8: Factors for Units Listed Alphabetically," [Online]. Available: <https://www.nist.gov/pml/special-publication-811/nist-guide-si-appendix-b-conversion-factors/nist-guide-si-appendix-b8>. [Accessed 20 September 2025].
- [6] NIST, "Ethanol," [Online]. Available: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C64175&Mask=2>. [Accessed 20 September 2025].
- [7] K. W. A. Guy, "The Explosive Decomposition of Nitrous Oxide," 1978. [Online]. Available: <https://www.ibb.ch/publication/N2O/Guy-KWA-1978-PhD-Thesis.pdf>. [Accessed 20 September 2025].

7.10 Appendix J: Thermal and Stress Modelling Python Code

Listing 1: main.py

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.interpolate import CubicSpline
4
5 from thermal import run_thermal_analysis
6 from stress import calc_stress
7 from materials import CuCrZr, AlSi10Mg, Inconel718, SS316
8
9 ### ----- GEOMETRY
10 -----
11 location_RPA = np.array([0, 8.95, 17.89, 26.84, 35.79, 44.73, 53.68,
12     62.63, 71.57, 80.52, 89.47, 98.41, 107.36, 116.31, 125.26, 134.2,
13     143.15, 149.51, 155.88, 162.24, 168.6, 174.97, 181.33, 187.69, 194.06,
14     200.42, 206.78, 213.15, 214.03, 226.76, 239.48, 252.21, 264.94, 277.66,
15     290.39, 310.93]) / 1000
16 inner_radius_RPA = np.array([48.33, 48.33, 48.33, 48.33, 48.33, 48.33,
17     48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33,
18     48.33, 48.05, 47.2, 45.76, 43.69, 40.93, 37.44, 33.76, 30.09, 26.82,
19     24.9, 24.18, 24.17, 27.25, 30.66, 34.07, 37.48, 40.89, 44.3, 49.8]) /
20     1000
21
22 location = np.linspace(location_RPA[0], location_RPA[-1], 1000)
23 inner_radius = CubicSpline(location_RPA, inner_radius_RPA)(location)
24
25 geometry = {
26     'location': location,
27     'inner_radius': inner_radius,
28 }
29
30 ### ----- CHANNEL
31 PARAMETERS -----
32 inner_wall_thickness = 0.001 #m
33 channel_height = 0.000924 #m
34 rib_width = 0.001 + 0.0001 #m
35 num_channels = 79
36 mass_flow_eth = 1.25 #kg/s
37 abs_roughness = 0.00003 #m
38
39 rho_ethanol = lambda T: -0.000964*T**2 - 0.3319*T + 970.1891
40 Cp_ethanol = lambda T: 0.035303*T**2 - 12.4495*T + 3020.93
41 visc_ethanol = lambda T: np.exp(0.000053*T**2 - 0.051588*T + 3.7854)
42 k_ethanol = lambda T: -0.000264*T + 0.2467
```

```

33 vapourP_ethanol = lambda T: 10**((8.04494 - (1554.3)/(222.65 + T - 273.15))
    * 133.322
34
35 width_channel      = (2 * np.pi * (inner_radius + inner_wall_thickness)) /
    num_channels - rib_width
36 area_channel       = width_channel * channel_height
37 velocity_channel    = lambda T, i: mass_flow_eth / (rho_ethanol(T) *
    area_channel[i]) / num_channels
38 hydraulic_diameter = 4 * area_channel / (2 * (width_channel +
    channel_height))
39 Re                  = lambda T, i: rho_ethanol(T) * velocity_channel(T, i)
    * hydraulic_diameter[i] / visc_ethanol(T)
40
41 channel_geom = {
42     'inner_wall_thickness': inner_wall_thickness,
43     'channel_height':      channel_height,
44     'rib_width':           rib_width,
45     'num_channels':        num_channels,
46     'mass_flow_eth':       mass_flow_eth,
47     'abs_roughness':       abs_roughness,
48     'k_cu':                CuCrZr['k'],
49     'width_channel':       width_channel,
50     'area_channel':        area_channel,
51     'velocity_channel':    velocity_channel,
52     'hydraulic_diameter':  hydraulic_diameter,
53     'Re':                  Re,
54 }
55
56 coolant_props = {
57     'rho': rho_ethanol,
58     'Cp':  Cp_ethanol,
59     'visc': visc_ethanol,
60     'k':    k_ethanol,
61     'vap':  vapourP_ethanol,
62 }
63
64 ### ----- GAS PARAMETERS
    -----
65 gas_params = {
66     'gamma': 1.2649,
67     'Tg_throat': 2102.76, #K
68     'P_throat': 1.5223e6, #Pa
69     'P_Chamber': 2.7579e6, #Pa
70     'Cp': 1.9846e3, #J/kgK

```

```

71     'visc_gas':      0.000069649, #Pa*s
72     'Pr':           0.5552,
73     'C_star':       1501.8,
74     'R_curvature':  36.25e-3,    #m
75 }
76
77 operating = {
78     'T_eth_inlet':    273.15 - 10,    #K
79     'P_coolant_inlet': 6.2e6,          #Pa
80     'TEOS_effect':    0.80,
81 }
82
83 ### ----- RUN ANALYSIS
84 -----
85
86 thermal = run_thermal_analysis(geometry, channel_geom, gas_params,
87                                coolant_props, operating)
88
89 print(f"Total Pressure Drop:      {thermal['total_dp']/1e6:.3f} MPa  ({
90     thermal['total_dp']/6894.76:.1f} psi)")
91 print(f"Pressure drop / Pc:      {thermal['total_dp']/gas_params['
92     P_Chamber']*100:.1f}%")
93 print(f"Peak wall temp (gas side): {np.max(thermal['Tw_gas']):.1f} K  ({np
94     .max(thermal['Tw_gas'])-273.15:.1f} degC)")
95 print(f"Ethanol exit temperature: {thermal['Te_all'][0]-273.15:.1f} degC"
96     )
97 print(f"Film Boil temp Throat:    {thermal['T_film_throat']:.1f} K")
98 print(f"Peak wall temp (eth side): {np.max(thermal['Tw_eth']):.1f} K")
99
100 materials = [CuCrZr, AlSi10Mg, Inconel718, SS316]
101
102 fos_results = {}
103 for mat in materials:
104     fos_op, fos_sd, fos_ht = calc_stress(mat, thermal, channel_geom,
105                                         gas_params['P_Chamber'],
106                                         inner_wall_thickness)
107     fos_results[mat['name']] = (fos_op, fos_sd, fos_ht)
108
109 ### ----- PLOTTING
110 -----
111
112 location      = thermal['location']
113 index_throat  = thermal['index_throat']
114
115 # Thermal plot
116 plt.figure(figsize=(10, 5))

```

```

107 plt.plot(location * 1000, thermal['Tw_gas'], label='Tw gas')
108 plt.plot(location * 1000, thermal['Tw_eth'], label='Tw eth')
109 plt.plot(location * 1000, thermal['Te_all'], label='Te coolant')
110 plt.axvline(location[index_throat] * 1000, color='gray', linestyle=':',
    label='Throat')
111 plt.xlabel('Location (mm)')
112 plt.ylabel('Temperature (K)')
113 plt.legend()
114 plt.grid(True)
115 plt.tight_layout()
116
117 # FOS plots
118 cases = [
119     (0, 'Operational Von Mises FOS'),
120     (1, 'Shutdown Von Mises FOS'),
121     (2, 'Hydrotest (1500 psi) Von Mises FOS'),
122 ]
123
124 for idx, title in cases:
125     fig, ax = plt.subplots(figsize=(12, 6))
126     for mat in materials:
127         if mat['name'] == "CuCrZr":
128             #ax.plot(location * 1000, fos_results[mat['name']][idx], label
    =mat['name'], marker='s', markevery=0.05)
129             ax.plot(location * 1000, fos_results[mat['name']][idx], label=
    mat['name'])
130         else:
131             ax.plot(location * 1000, fos_results[mat['name']][idx], label=
    mat['name'])
132     ax.axhline(1.5, color='red', linestyle='--', linewidth=1, label='FOS =
    1.5')
133     ax.axvline(location[index_throat] * 1000, color='gray', linestyle=':',
    label='Throat')
134     ax.set_xlabel('Location (mm)')
135     ax.set_ylabel('Factor of Safety')
136     ax.set_title(title)
137     ax.set_xlim([0, 308])
138     ax.legend()
139     if title == "Hydrotest (1500 psi) Von Mises FOS":
140         ax.set_ylim([0, 100])
141     else:
142         ax.set_ylim([0, 10])
143
144     ax.grid(True)

```

```

145     plt.tight_layout()
146
147 plt.show()

```

Listing 2: materials.py [13]

```

1  import numpy as np
2
3  # Temperature in K, E in Pa, Ys in Pa, alpha in 1/K, k in W/mK
4
5  CuCrZr = {
6      'name': 'CuCrZr',
7      'k': 350,
8      'alpha': 16e-6,
9      'T_E': np.array([25, 50, 100, 150, 200, 250, 300, 350, 400, 450,
10                     500, 550, 600, 650, 700]) + 273.15,
11      'E': np.array([129, 128, 126, 124, 122, 119, 116, 113, 110, 106,
12                    103, 99, 95, 90, 86]) * 1e9,
13      'T_Ys': np.array([25, 50, 100, 150, 200, 250, 300, 350, 400, 450,
14                      500, 550, 600, 650, 700]) + 273.15,
15      'Ys': np.array([290, 280, 275, 265, 255, 245, 235, 220, 200, 180,
16                     165, 145, 130, 100, 85]) * 1e6,
17  }
18
19  AlSi10Mg = {
20      'name': 'AlSi10Mg',
21      'k': 170,
22      'alpha': 27e-6,
23      'T_E': np.array([25, 50, 100, 150, 200, 250, 300, 350, 400, 450]) +
24             273.15,
25      'E': np.array([77.6, 75.5, 72.8, 71.0, 70.0, 65.0, 55.0, 41.0,
26                    35.0, 0.1]) * 1e9,
27      'T_Ys': np.array([25, 50, 100, 150, 200, 250, 300, 350, 400, 450]) +
28             273.15,
29      'Ys': np.array([204, 198, 181, 182, 158, 132, 70, 30, 12, 0.1]) * 1
30             e6,
31  }
32
33  Inconel718 = {
34      'name': 'Inconel 718',
35      'k': 11.4,
36      'alpha': 16e-6,
37      'T_E': np.array([21, 540, 650, 750, 870]) + 273.15,
38      'E': np.array([199.95, 171.07, 162.86, 153.32, 142.31]) * 1e9,
39      'T_Ys': np.array([21, 540, 650, 750, 870]) + 273.15,

```

```

32     'Ys':      np.array([1190, 1060, 1020, 740, 330]) * 1e6,
33 }
34
35 SS316 = {
36     'name':    'SS 316',
37     'k':       20,
38     'alpha':   20e-6,
39     'T_E':     np.array([27, 149, 260, 371, 482, 593, 704, 816]) + 273.15,
40     'E':       np.array([193, 190, 181, 172, 162, 153, 143, 132]) * 1e9,
41     'T_Ys':    np.array([27, 149, 260, 371, 482, 593, 704, 816]) + 273.15,
42     'Ys':      np.array([290, 201, 172, 159, 148, 140, 131, 110]) * 1e6,
43 }

```

Listing 3: optimization.py

```

1  # optimization.py
2
3  import numpy as np
4  import matplotlib.pyplot as plt
5  from scipy.interpolate import CubicSpline
6
7  from thermal import run_thermal_analysis
8  from materials import CuCrZr
9
10  """ ----- GEOMETRY
11  -----
12  location_RPA = np.array([0, 8.95, 17.89, 26.84, 35.79, 44.73, 53.68,
13  62.63, 71.57, 80.52, 89.47, 98.41, 107.36, 116.31, 125.26, 134.2,
14  143.15, 149.51, 155.88, 162.24, 168.6, 174.97, 181.33, 187.69, 194.06,
15  200.42, 206.78, 213.15, 214.03, 226.76, 239.48, 252.21, 264.94, 277.66,
16  290.39, 310.93]) / 1000
17  inner_radius_RPA = np.array([48.33, 48.33, 48.33, 48.33, 48.33, 48.33,
18  48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33, 48.33,
19  48.33, 48.05, 47.2, 45.76, 43.69, 40.93, 37.44, 33.76, 30.09, 26.82,
20  24.9, 24.18, 24.17, 27.25, 30.66, 34.07, 37.48, 40.89, 44.3, 49.8]) /
21  1000
22
23  location      = np.linspace(location_RPA[0], location_RPA[-1], 1000)
24  inner_radius  = CubicSpline(location_RPA, inner_radius_RPA)(location)
25
26  geometry = {
27     'location':    location,
28     'inner_radius': inner_radius,
29 }

```

```

22  ### ----- CONSTANT
    PARAMETERS -----
23  inner_wall_thickness = 0.001
24  rib_width            = 0.001 + 0.0001
25  mass_flow_eth        = 1.25
26  abs_roughness        = 0.00003
27
28  rho_ethanol = lambda T: -0.000964*T**2 - 0.3319*T + 970.1891
29  Cp_ethanol  = lambda T: 0.035303*T**2 - 12.4495*T + 3020.93
30  visc_ethanol = lambda T: np.exp(0.000053*T**2 - 0.051588*T + 3.7854)
31  k_ethanol    = lambda T: -0.000264*T + 0.2467
32  vapourP_ethanol = lambda T: 10**(8.04494 - (1554.3)/(222.65 + T - 273.15))
    * 133.322
33
34  coolant_props = {
35      'rho': rho_ethanol,
36      'Cp': Cp_ethanol,
37      'visc': visc_ethanol,
38      'k': k_ethanol,
39      'vap': vapourP_ethanol,
40  }
41
42  ### ----- GAS PARAMETERS
    -----
43  gas_params = {
44      'gamma': 1.2649,
45      'Tg_throat': 2102.76,
46      'P_throat': 1.5223e6,
47      'P_Chamber': 2.7579e6,
48      'Cp': 1.9846e3,
49      'visc_gas': 0.000069649,
50      'Pr': 0.5552,
51      'C_star': 1501.8,
52      'R_curvature': 36.25e-3,
53  }
54
55  operating = {
56      'T_eth_inlet': 273.15 - 10,
57      'P_coolant_inlet': 6.2e6,
58      'TEOS_effect': 0.80,
59  }
60
61  ### ----- BASELINE CONFIG (
    main.py) -----

```

```

62 baseline_channel_height = 0.000924
63 baseline_num_channels   = 79
64
65 width_channel_baseline = (2 * np.pi * (inner_radius + inner_wall_thickness
66     )) / baseline_num_channels - rib_width
67 area_channel_baseline  = width_channel_baseline * baseline_channel_height
68 hydraulic_diameter_baseline = 4 * area_channel_baseline / (2 * (
69     width_channel_baseline + baseline_channel_height))
70
71 velocity_channel_baseline = lambda T, i: mass_flow_eth / (rho_ethanol(T) *
72     area_channel_baseline[i]) / baseline_num_channels
73 Re_baseline = lambda T, i: rho_ethanol(T) * velocity_channel_baseline(T, i
74     ) * hydraulic_diameter_baseline[i] / visc_ethanol(T)
75
76 channel_geom_baseline = {
77     'inner_wall_thickness': inner_wall_thickness,
78     'channel_height':      baseline_channel_height,
79     'rib_width':           rib_width,
80     'num_channels':        baseline_num_channels,
81     'mass_flow_eth':       mass_flow_eth,
82     'abs_roughness':       abs_roughness,
83     'k_cu':                CuCrZr['k'],
84     'width_channel':       width_channel_baseline,
85     'area_channel':        area_channel_baseline,
86     'velocity_channel':    velocity_channel_baseline,
87     'hydraulic_diameter':  hydraulic_diameter_baseline,
88     'Re':                  Re_baseline,
89 }
90
91 baseline_thermal = run_thermal_analysis(
92     geometry,
93     channel_geom_baseline,
94     gas_params,
95     coolant_props,
96     operating
97 )
98
99 baseline_dp = baseline_thermal['total_dp']
100 baseline_Tw = np.max(baseline_thermal['Tw_gas'])
101
102 ### ----- OPTIMIZATION
103     RANGES -----
104 channel_heights = np.linspace(0.001, 0.005, 25)
105 channel_counts  = np.arange(20, 80, 5)

```

```

101
102 pressure_drop_results = []
103 wall_temp_results     = []
104
105 ### ----- OPTIMIZATION LOOP
106 -----
107 for channel_height in channel_heights:
108     for num_channels in channel_counts:
109
110         width_channel      = (2 * np.pi * (inner_radius +
111             inner_wall_thickness)) / num_channels - rib_width
112         area_channel       = width_channel * channel_height
113         hydraulic_diameter = 4 * area_channel / (2 * (width_channel +
114             channel_height))
115
116         velocity_channel = lambda T, i: mass_flow_eth / (rho_ethanol(T) *
117             area_channel[i]) / num_channels
118         Re = lambda T, i: rho_ethanol(T) * velocity_channel(T, i) *
119             hydraulic_diameter[i] / visc_ethanol(T)
120
121         channel_geom = {
122             'inner_wall_thickness': inner_wall_thickness,
123             'channel_height':      channel_height,
124             'rib_width':           rib_width,
125             'num_channels':        num_channels,
126             'mass_flow_eth':       mass_flow_eth,
127             'abs_roughness':       abs_roughness,
128             'k_cu':                CuCrZr['k'],
129             'width_channel':       width_channel,
130             'area_channel':        area_channel,
131             'velocity_channel':    velocity_channel,
132             'hydraulic_diameter':  hydraulic_diameter,
133             'Re':                  Re,
134         }
135     }
136     print("cycle completed")
137     try:
138         thermal = run_thermal_analysis(
139             geometry,
140             channel_geom,
141             gas_params,
142             coolant_props,
143             operating
144         )
145

```

```

140         pressure_drop_results.append(thermal['total_dp'])
141         wall_temp_results.append(np.max(thermal['Tw_gas']))
142
143     except Exception:
144         continue
145
146     ### ----- RESULTS PLOT
147     -----
148     pressure_drop_results = np.array(pressure_drop_results)
149     wall_temp_results      = np.array(wall_temp_results)
150
151     plt.figure(figsize=(10,6))
152
153     plt.scatter(
154         wall_temp_results,
155         pressure_drop_results/1e6,
156         s=20,
157         label="Optimization Trials",
158         alpha = 0.5
159     )
160
161     plt.scatter(
162         baseline_Tw,
163         baseline_dp/1e6,
164         s=250,
165         marker="*",
166         color="red",
167         label="Printed Configuration"
168     )
169
170     plt.xlabel("Max Gas-Side Wall Temperature (K)")
171     plt.ylabel("Pressure Drop (MPa)")
172     plt.title("Cooling Channel Optimization Trials")
173     plt.legend()
174     plt.grid(True)
175     plt.tight_layout()
176
177     plt.show()

```

Listing 4: stress.py [5]

```

1 import numpy as np
2
3

```

```

4 def calc_stress(material, thermal_results, channel_geom, P_Chamber, t, v
   =0.33):
5     name      = material['name']
6     alpha_w   = material['alpha']
7     k_w       = material['k']
8
9     location   = thermal_results['location']
10    inner_radius = thermal_results['inner_radius']
11    Tw_gas      = thermal_results['Tw_gas']
12    Tw_eth      = thermal_results['Tw_eth']
13    q_gas       = thermal_results['q_gas']
14    P_coolant_dist = thermal_results['P_coolant_dist']
15    num_channels = channel_geom['num_channels']
16    rib_width    = channel_geom['rib_width']
17
18    E_w = np.interp(Tw_gas, material['T_E'], material['E'])
19    Ys_w = np.interp(Tw_gas, material['T_Ys'], material['Ys'])
20
21    Do      = (inner_radius + t) * 2
22    cw      = Do * np.pi / num_channels - rib_width
23    Di      = inner_radius * 2
24    P_hydrotest = 1500 * 6894.76 # 1500 psi in Pa
25
26    sigma_l = E_w * alpha_w * (Tw_gas - Tw_eth)
27    sigma_tt = (E_w * alpha_w * q_gas * t) / (2 * (1 - v) * k_w)
28    sigma_buck = (E_w * E_w * (t / Di)) / ((np.sqrt(E_w) + np.sqrt(E_w))
   **2 * np.sqrt(3 * (1 - v**2)))
29
30    def von_mises(sigma_t, sigma_l):
31        return np.sqrt(0.5 * ((sigma_t - sigma_l)**2 + sigma_l**2 +
   sigma_t**2))
32
33    def fos_and_print(label, sigma_t, sigma_l_case, Ys):
34        sigma_vm = von_mises(sigma_t, sigma_l_case)
35        fos      = Ys / sigma_vm
36        buck_ratio = sigma_l_case / sigma_buck
37        print(f" {label}:")
38        print(f"      Min FOS: {np.min(fos):.2f} at x={location[
   np.argmin(fos)]*1000:.1f} mm")
39        print(f"      Max Von Mises: {np.max(sigma_vm)/1e6:.1f} MPa")
40        print(f"      Max Buckling Ratio: {np.max(buck_ratio):.3f} (should
   be < 0.9)")
41        return fos
42

```

```

43     print(f"\n{'='*50}")
44     print(f"    Material: {name}")
45     print(f"{'='*50}")
46
47     # Operational case
48     sigma_tp_op = ((P_coolant_dist - P_Chamber) / 2) * (cw / t)**2
49     sigma_t_op  = sigma_tp_op + sigma_tt
50     fos_op      = fos_and_print("Operational", sigma_t_op, sigma_l, Ys_w)
51
52     # Shutdown case
53     sigma_tp_sd = (P_coolant_dist / 2) * (cw / t)**2
54     sigma_t_sd  = sigma_tp_sd + sigma_tt
55     fos_sd      = fos_and_print("Shutdown", sigma_t_sd, sigma_l, Ys_w)
56
57     # Hydrotest case (cold material, no combustion, no heat flux)
58     Ys_cold     = np.interp(293.15, material['T_Ys'], material['Ys'])
59     sigma_tp_ht = (P_hydrotest / 2) * (cw / t)**2
60     sigma_vm_ht = von_mises(sigma_tp_ht, np.zeros(len(location)))
61     fos_ht      = Ys_cold / sigma_vm_ht
62     print(f"    Hydrotest (1500 psi):")
63     print(f"        Min FOS:                {np.min(fos_ht):.2f}    at x={location[
        np.argmin(fos_ht)]*1000:.1f} mm")
64     print(f"        Max Von Mises:            {np.max(sigma_vm_ht)/1e6:.1f} MPa")
65
66     return fos_op, fos_sd, fos_ht

```

Listing 5: thermal.py

```

1  import numpy as np
2  from scipy.optimize import brentq
3  from scipy.optimize import fsolve
4
5
6  def mach_solver(AR, gamma, location):
7      index_throat = np.argmin(np.abs(AR - 1))
8      Mach         = np.zeros(len(location))
9      area_ratio   = lambda M: (1/M) * ((2/(gamma+1)) * (1 + (gamma-1)/2 * M
      **2))**((gamma+1)/(2*(gamma-1)))
10     for i in range(len(AR)):
11         if i < index_throat:
12             Mach[i] = brentq(lambda M: area_ratio(M) - AR[i], 1e-6, 1.0 -
                1e-9)
13         elif i == index_throat:
14             Mach[i] = 1
15         else:

```

```

16         Mach[i] = brentq(lambda M: area_ratio(M) - AR[i], 1.0 + 1e-9,
17                             50.0)
18     return Mach
19
20 def bartz(Tw, Tg_t, M, AR, gas_params, R_throat):
21     gamma = gas_params['gamma']
22     visc_gas = gas_params['visc_gas']
23     Cp = gas_params['Cp']
24     Pr = gas_params['Pr']
25     P_Chamber = gas_params['P_Chamber']
26     C_star = gas_params['C_star']
27     R_curvature = gas_params['R_curvature']
28     D_t = R_throat * 2
29     sigma = (0.5 * (Tw / Tg_t) * (1 + (gamma - 1) / 2 * M**2) + 0.5)
30             **(-0.68) \
31             * (1 + (gamma - 1) / 2 * M**2)**(-0.12)
32     hg = (0.026 / D_t**0.2) * (visc_gas**0.2 * Cp / Pr**0.6) \
33         * (P_Chamber / C_star)**0.8 * (D_t / R_curvature)**0.1 *
34         AR**(-0.9) * sigma
35
36     return hg
37
38 def friction_factor(Re_loc, dh, abs_roughness):
39     return (1 / (-1.8 * np.log10((6.9 / Re_loc) + (abs_roughness / dh /
40         3.7)**1.11))))**2
41
42 def h_channel(T_e, i, channel_geom, coolant_props, abs_roughness):
43     width_channel = channel_geom['width_channel']
44     hydraulic_diameter = channel_geom['hydraulic_diameter']
45     channel_height = channel_geom['channel_height']
46     rib_width = channel_geom['rib_width']
47     k_cu = channel_geom['k_cu']
48     Re = channel_geom['Re']
49     Cp_ethanol = coolant_props['Cp']
50     visc_ethanol = coolant_props['visc']
51     k_ethanol = coolant_props['k']
52
53     Pr_local = Cp_ethanol(T_e) * visc_ethanol(T_e) / k_ethanol(T_e)
54     Re_loc = Re(T_e, i)
55     f = friction_factor(Re_loc, hydraulic_diameter[i],
56         abs_roughness)

```

```

54     Nu        = (f / 8) * (Re_loc - 1000) * Pr_local / (1 + 12.7 * (f / 8)
        **0.5 * (Pr_local**(2/3) - 1))
55     h         = Nu * k_ethanol(T_e) / hydraulic_diameter[i]
56     m         = (2 * h / k_cu / rib_width)**0.5
57     n_fin     = np.tanh(m * channel_height) / (m * channel_height)
58     A_total   = width_channel[i] + 2 * channel_height
59     A_fin     = 2 * channel_height
60     n_total   = 1 - (A_fin / A_total) * (1 - n_fin)
61     return n_total * h
62
63
64 def run_thermal_analysis(geometry, channel_geom, gas_params, coolant_props
    , operating):
65     location      = geometry['location']
66     inner_radius  = geometry['inner_radius']
67     width_channel = channel_geom['width_channel']
68     hydraulic_diameter = channel_geom['hydraulic_diameter']
69     inner_wall_thickness = channel_geom['inner_wall_thickness']
70     k_cu          = channel_geom['k_cu']
71     num_channels  = channel_geom['num_channels']
72     mass_flow_eth = channel_geom['mass_flow_eth']
73     abs_roughness = channel_geom['abs_roughness']
74     Re            = channel_geom['Re']
75     rho_ethanol   = coolant_props['rho']
76     Cp_ethanol    = coolant_props['Cp']
77     vapourP_ethanol = coolant_props['vap']
78     velocity_channel = channel_geom['velocity_channel']
79     gamma         = gas_params['gamma']
80     Pr            = gas_params['Pr']
81
82     A_throat      = np.pi * np.min(inner_radius)**2
83     A_all         = np.pi * inner_radius**2
84     AR            = A_all / A_throat
85     index_throat  = np.argmin(np.abs(AR - 1))
86     R_throat      = inner_radius[index_throat]
87
88     # Mach and gas temperature distribution
89     M_all         = mach_solver(AR, gamma, location)
90     Tg_t          = gas_params['Tg_throat'] * (1 + (gamma - 1) / 2)
91     Tg_all        = Tg_t / (1 + (gamma - 1) / 2 * M_all**2)
92     r             = Pr**(1/3)
93     T_aw          = Tg_all * (1 + r * (gamma - 1) / 2 * M_all**2)
94

```

```

95     P_t    = gas_params['P_throat'] / ((1 + (gamma - 1) / 2)**(-gamma / (
        gamma - 1)))
96     P_all = ((1 + (gamma - 1) / 2 * M_all**2)**(-gamma / (gamma - 1))) *
        P_t
97
98     # Heat transfer analysis
99     T_eth_inlet = operating['T_eth_inlet']
100    Tw_gas = np.full(len(location), 273.0)
101    Tw_eth = np.full(len(location), 273.0)
102    Te_all = np.full(len(location), T_eth_inlet)
103    h_gas  = np.zeros(len(location))
104    q_gas  = np.zeros(len(location))
105    q_eth  = np.zeros(len(location))
106    h_eth  = np.zeros(len(location))
107
108    for i in range(len(location)-1, 0, -1):
109        Tw_gas[i-1] = Tw_gas[i]
110        q_diff = 10
111        while abs(q_diff) > 0.01:
112            h_gas[i-1] = bartz(Tw_gas[i-1], Tg_t, M_all[i-1], AR[i-1],
                gas_params, R_throat)
113            q_gas[i-1] = h_gas[i-1] * (T_aw[i-1] - Tw_gas[i-1]) *
                operating['TEOS_effect']
114            Tw_eth[i-1] = Tw_gas[i-1] - q_gas[i-1] * inner_wall_thickness
                / k_cu
115            h_eth[i-1] = h_channel(Te_all[i], i-1, channel_geom,
                coolant_props, abs_roughness)
116            q_eth[i-1] = h_eth[i-1] * (Tw_eth[i-1] - Te_all[i])
117            q_diff      = q_eth[i-1] - q_gas[i-1]
118            Tw_gas[i-1] -= q_diff / (h_gas[i-1] + h_eth[i-1])
119            Te_all[i-1] = Te_all[i] + q_gas[i-1] * width_channel[i-1] * (
                location[i] - location[i-1]) \
120                / (mass_flow_eth / num_channels * Cp_ethanol(Te_all[
                    i]))
121
122    Tw_gas[-1] = Tw_gas[-2]
123    Tw_eth[-1] = Tw_eth[-2]
124
125    # Pressure drop
126    pressure_drop_values = []
127    for i in range(len(location) - 1):
128        v_loc    = velocity_channel(Te_all[i], i)
129        Re_loc   = Re(Te_all[i], i)

```

```

130     ff      = friction_factor(Re_loc, hydraulic_diameter[i],
        abs_roughness)
131     dx      = location[i+1] - location[i]
132     delta_p = ff * (dx / hydraulic_diameter[i]) * (0.5 * rho_ethanol(
        Te_all[i]) * v_loc**2)
133     pressure_drop_values.append(delta_p)
134
135     cumulative_dp = np.cumsum(pressure_drop_values)
136     total_dp      = sum(pressure_drop_values)
137
138     P_coolant_inlet      = operating['P_coolant_inlet']
139     P_coolant_dist       = np.zeros(len(location))
140     P_coolant_dist[0]    = P_coolant_inlet - total_dp
141     P_coolant_dist[1:]   = P_coolant_inlet - (total_dp - cumulative_dp)
142
143     T_ethanol_film = lambda P: fsolve(lambda T: vapourP_ethanol(T) - P, x0
        =350)[0]
144     T_film_throat = T_ethanol_film(P_coolant_inlet - P_coolant_dist[
        index_throat])
145     return {
146         'location':      location,
147         'inner_radius':  inner_radius,
148         'index_throat':  index_throat,
149         'Tw_gas':        Tw_gas,
150         'Tw_eth':        Tw_eth,
151         'Te_all':        Te_all,
152         'q_gas':         q_gas,
153         'h_gas':         h_gas,
154         'h_eth':         h_eth,
155         'T_aw':          T_aw,
156         'M_all':         M_all,
157         'AR':            AR,
158         'P_coolant_dist': P_coolant_dist,
159         'total_dp':      total_dp,
160         'T_film_throat': T_film_throat,
161     }

```

Listing 6: TankClosure.m

```

1 clear
2 close
3
4 % Aluminum Properties https://asm.matweb.com/search/specificmaterial.asp?
    bassnum=ma6061t6
5 yield_al = 40000; %psi

```

```

6 shear_al = yield_al / sqrt(3);
7 bearing_al = 56000; %psi
8
9 % Bolt Properties https://www.mcmaster.com/products/button-head-socket-cap-screws/drive-style~hex/material~alloy-steel/finish~black-oxide/rounded-head-profile~standard/thread-size~5-16-24-2/
10 % assuming M8 bolt
11 UTS_bolt = 120000; %psi
12 USS_bolt = UTS_bolt/sqrt(3); %psi
13 min_diam = 0.2617; %in
14 shear_area = pi * (min_diam / 2)^2;
15
16 % Tank Properties
17 MEOP = 1000; %Psi
18 tank_OD = 8; %in
19 wall_thickness = 0.2; %in
20 tank_ID = tank_OD - 2*wall_thickness;
21 force = pi * (tank_ID / 2)^2 * MEOP;
22
23 % Variable Setups
24
25 num_bolts = 30:31;
26 edge_distance = [0.560, 0.5601];
27
28 % num_bolts = 24;
29 % edge_distance = 0.56;
30
31 %Matrix Setup: xdim: number of bolts, ydim: edge_distance, zdim: FOS of
32 %failure modes
33
34 result_all = zeros(numel(num_bolts), numel(edge_distance), 4);
35 worst_FOS = zeros(numel(num_bolts), numel(edge_distance));
36
37 for bolt_index = 1:numel(num_bolts)
38     for edge_index = 1:numel(edge_distance)
39         result_all(bolt_index, edge_index, :) = failureCalc(yield_al,
40             shear_al, bearing_al, USS_bolt, shear_area, wall_thickness,
41             force, num_bolts(bolt_index), edge_distance(edge_index));
42         worst_FOS(bolt_index, edge_index) = min(result_all(bolt_index,
43             edge_index, :));
44     end
45 end
46
47 surf(worst_FOS)

```

```

45 ylabel("Number of Bolts")
46 xlabel("Edge Distance from Edge of Bolt Hole [in]")
47 zlabel("Worst FOS")
48 original_ticks = xticks;
49 new_labels = original_ticks / 100;
50 xticklabels(new_labels);
51
52 %% Failure mode calculation
53
54 function FOS = failureCalc(yield_al, shear_al, bearing_al, USS_bolt,
    shear_area, wall_thickness, force, num_bolts, edge_distance)
55     %bolt_shear
56     force_bolt = force / num_bolts;
57     bolt_shear_FOS = USS_bolt / (force_bolt / shear_area);
58     %bearing
59     bearing_area = wall_thickness * 0.314961;
60     bearing_FOS = bearing_al / (force_bolt / bearing_area);
61     %bulk failure
62     bulk_area = pi * (4^2 - (4-wall_thickness)^2) - 0.354331 *
        wall_thickness * num_bolts;
63     bulk_FOS = yield_al / (force / bulk_area);
64     %tearout
65     tear_area = 0.354331 * wall_thickness * 2;
66     tear_FOS = shear_al / (force_bolt / tear_area);
67
68     FOS = [bolt_shear_FOS, bearing_FOS, bulk_FOS, tear_FOS];
69 end

```

7.11 Appendix K: Thrust and Chamber Pressure Estimation Script

Listing 7: main.py

```

1 import CoolProp as cp
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5 import math
6
7 fluid = 'NitrousOxide'
8 psi_to_pa = 6894.7572931783
9
10 fuel_cdA = 0.0000142551898
11 ox_cdA = 0.00002845677614
12 mdot = 2.8 #kg/s
13
14 ethanol_density = 789 #kg/m^3

```

```

15
16 df = pd.read_csv(r"C:\Users\mgord\OneDrive\Desktop\Waterloo\FYDP\Fire Data
    .csv")
17
18 Temp = 273.15-15
19 Pvap = cp.CoolProp.PropsSI('P', 'T', Temp, 'Q', 0, fluid)
20 print("Solving...")
21
22 for index in df.index:
23
24     ox_press = df.loc[index,"OPT-301 (Ox Tank) (psi)"]*psi_to_pa
25     fuel_press = df.loc[index, "FPT-301 (Fuel Tank) (psi)"]*psi_to_pa
26     p_guess = 300
27     mdot_guess = 0
28     cc_pressures = np.linspace(200,fuel_press/psi_to_pa,num=int((fuel_press
        /psi_to_pa-200))
29     h1 = cp.CoolProp.PropsSI('H','T', Temp,'P',ox_press, fluid)
30     s1 = cp.CoolProp.PropsSI('S','T',Temp,'P',ox_press, fluid)
31     ox_hem_max = 0
32     ox_hem_max_p = 0
33     for cc_press in cc_pressures:
34         cc_pressure = cc_press*psi_to_pa
35         s2l = cp.CoolProp.PropsSI('S','P', cc_pressure,'Q',0, fluid)
36         s2v = cp.CoolProp.PropsSI('S','P', cc_pressure,'Q',1, fluid)
37         x = (s1-s2l)/(s2v-s2l)
38         if 0 < x and x < 1:
39             h2 = cp.CoolProp.PropsSI('H','P',cc_pressure, 'Q', x, fluid)
40             rho2 = cp.CoolProp.PropsSI('D','P',cc_pressure, 'Q', x, fluid)
41         else:
42             h2 = cp.CoolProp.PropsSI('H','P',cc_pressure, 'S', s1, fluid)
43             rho2 = cp.CoolProp.PropsSI('D','P',cc_pressure, 'S', s1, fluid
                )
44         if h2 < h1:
45             ox_hem = rho2*np.sqrt(2*(h1-h2))
46         else:
47             ox_hem = 0
48         if ox_hem > ox_hem_max:
49             ox_hem_max = ox_hem
50             ox_hem_max_p = cc_pressure
51     while abs(mdot-mdot_guess)>0.01:
52         cc_pressure = p_guess*psi_to_pa
53         fuel_mdot = fuel_cdA*np.sqrt(2*ethanol_density*(fuel_press-
            cc_pressure))

```

```

54     ox_spi = ox_cdA*np.sqrt(2*cp.CoolProp.PropsSI('D', 'P', ox_press,
55         'T', Temp, fluid)*(ox_press-cc_pressure))
56     s2l = cp.CoolProp.PropsSI('S', 'P', cc_pressure, 'Q', 0, fluid)
57     s2v = cp.CoolProp.PropsSI('S', 'P', cc_pressure, 'Q', 1, fluid)
58     x = (s1-s2l)/(s2v-s2l)
59     if 0 < x and x < 1:
60         h2 = cp.CoolProp.PropsSI('H', 'P', cc_pressure, 'Q', x, fluid)
61         rho2 = cp.CoolProp.PropsSI('D', 'P', cc_pressure, 'Q', x, fluid)
62     else:
63         h2 = cp.CoolProp.PropsSI('H', 'P', cc_pressure, 'S', s1, fluid)
64         rho2 = cp.CoolProp.PropsSI('D', 'P', cc_pressure, 'S', s1, fluid)
65     if h2 < h1:
66         ox_hem = rho2*np.sqrt(2*(h1-h2))
67     if ox_hem < ox_hem_max and cc_pressure < ox_hem_max_p:
68         ox_hem = ox_hem_max
69     ox_hem = ox_cdA*ox_hem
70     if Pvap > cc_pressure:
71         k = np.sqrt((ox_press-cc_pressure)/(Pvap-cc_pressure))
72         ox_dyer = (k/(1+k)*ox_spi+1/(1+k)*ox_hem)
73         mdot_guess = ox_dyer+fuel_mdot
74     else:
75         mdot_guess = ox_spi+fuel_mdot
76     p_guess = p_guess*np.sqrt(mdot_guess/mdot)
77     df.loc[index, "CC Pressure"] = p_guess
78     df.loc[index, "Thrust"] = p_guess*psi_to_pa*np.pi*0.0274**2
79     df.loc[index, "OF"] = (mdot_guess-fuel_mdot)/fuel_mdot
80     print("\033[A\033[A")
81     print('Solving {:.2f}%'.format(df.loc[index, "Test Time"]/max(df["Test
82         Time"]*100)))
83
84
85 fig, ax = plt.subplots(nrows = 1, ncols = 1)
86
87 ax.plot(df["Test Time"], df["OPT-301 (Ox Tank) (psi)"], label = "Ox Tank
88     Pressure")
89 ax.plot(df["Test Time"], df["FPT-301 (Fuel Tank) (psi)"], label = "Fuel
90     Tank Pressure")
91 ax.plot(df["Test Time"], df["CC Pressure"], label = "Estimated CC Pressure"
92     )
93 ax.set_ylabel("Pressure (psi)")
94 ax.set_xlabel("Burn time")
95 fig.legend()
96 fig.show()

```

```

92 input()
93
94 fig1, ax1 = plt.subplots(nrows = 1, ncols = 1)
95
96
97 ax2 = ax1.twinx()
98 ax1.plot(df["Test Time"], df["OF"], color='red',label = "OF Ratio",)
99 ax2.plot(df["Test Time"],df["Thrust"]/1000, color = 'blue',label = "Thrust
   ")
100 ax1.set_ylabel("Oxidizer to Fuel Mass Ratio")
101 ax2.set_ylabel("Thrust [kN]")
102 ax1.set_xlabel("Burn time")
103 fig1.legend()
104 fig1.show()
105 input()
106
107 df.to_csv(r"C:\Users\mgord\OneDrive\Desktop\Waterloo\FYDP\Fire Data.csv")

```